Course Name: Computer Network

Course Outcome

CO1: Understand the concept of OSI and TCP/IP reference model
CO2: Understand the basics of data transmission at physical layer.
CO3: Understand the channel allocation using ALOHA, CSMA and CSMA/CD.
CO4: Apply error detection and correction technique to eliminate transmission error.
CO5: Analyze the fixed and variable length address (IPv4) subneting for the given scenarios.
CO6: Understand the design issues of the transport layer.
CO7: Understand the mechanism of protocols at application layer such as FTP, HTTP, Telnet, DNS.
CO8: Understand IPv6 addressing and differentiate it from IPv4.

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End Term Examination, Even Semester 2021-22
B.Tech (CSE), II Year, IV Semester
BCSC0008 Computer Network

Time: 3 Hours **Maximum Marks: 50**

Section – A

Attempt All Questions 4 X 5 = 20 Marks

No.	Detail of Question	Marks	CO	BL	KL
1	Given the data word 10100111 and the divisor 10111.Show the generation of the codeword at the sender site (using binary division). Show the checking of the codeword at the receiver site (assume no error).	4	CO4	An	P
2	Compare Go-Back-N protocol with Selective-Repeat protocol in terms of efficiency, number of retransmission and window size (both sender and receiver end).	4	CO3	U	C
3	What are the responsibilities of Transport layer and Data Link layer in OSI reference model? Why the error control and flow control mechanism used in both layer of OSI model?	4	CO6	R	C
4	An Internet Service Provider (ISP) has the following chunk of CIDR-based IP addresses available with it: 245.248.128.0/20. The ISP wants to give half of this chunk of addresses to Organization A, and a quarter to Organization B, while retaining the remaining with itself. Write valid allocation of addresses to A and B?	4	CO5	A	P
5	The only entries in a certain route table are (128.59.28.0/22, port 0), (128.59.28.0/23, port 1) and (128.59.28.0/24, port 2). These entries indicate CIDR network number, the prefix	4	CO5	A	P

	indication and the corresponding port to which a packet should be forwarded. If a packet arrives with a destination IP address equal to 128.59.29.18, which port will this router will this router forward the packet to? OR A Datagram of 3000 B (20 B header +2980 B Data) reached at router and must be forwarded to a link with MTU of 500 B. how many fragmentations will be generated and also write MF (more fragment), offset and total length value for all.				

Section – B

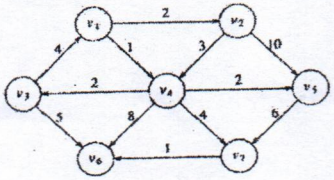
Attempt All Questions 3 X 5 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
1	Suppose an ISP owns the block of addresses of the form 101.101.128/17. Suppose it wants to create four subnets from this block, with each block having the same number of IP addresses. What are the prefixes (of form a.b.c.d/x) and Network address for the four subnets?	3	CO5	A	P
2	What do you mean by open loop and closed loop congestion control policies? Consider the effect of using slow start on a line with a 10 ms RTT and no congestion. The receiver window is 24 KB and the maximum segment size is 2 KB. How long does it take before the first full window can be sent?	3	CO6	U	C
3	Discuss the advantages of IPv6 addressing scheme over IPv4. Discuss the header format of IPv4.	3	CO8	R	P
4	The following is the dump of a TCP header in hexadecimal format: 05320017 00000001 00000000 500207FF 00000000 a) What is the source port number? b) What is the destination port number? c) What is the sequence number? d) What is the acknowledgement number? e) What is the window size? f) What is the segment type?	3	CO6	A	C
5	What is the function of SMTP? Differentiate	3	CO7	U	P

	between a User Agent (UA) and a Mail Transfer Agent. How does MIME enhance SMTP?				

Section – C

*Attempt **All** Questions* 5 X 3 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
1	Explain the concept of link state routing. Using link state routing algorithm on the given network having seven routers. Find the routing table for router v1? 	5	CO5	A	P
2	Describe three-way handshaking protocol used by TCP to establish a connection. Also discuss the SYN Flooding Attack, which is a serious security problem for connection establishment of TCP.	5	CO6	U	C
3	Explain the concept and applications of Public Key Cryptosystem and write the Key generation algorithm of RSA. Also generate the public key and secret key for the following prime numbers using RSA algorithm: P = 7, Q= 11 and assume e=7.	5	CO7	A	C

CO – Course Outcome, BL – Abbreviation for Bloom's Taxonomy Level (R-Remember, U-Understand, A-Apply, An-Analyze, E-Evaluate, C-Create), KL – Abbreviation for Knowledge Level (F-Factual, C-Conceptual, P-Procedural, M-Metacognitive). However, For Engg. Courses in addition to F, C, P & M include D-Fundamental Design Principles, S-Criteria and Specifications, PC-Practical Constraints, DI- Design Instrumentalities

Course Name: Engineering Mathematics III

Course Outcomes:

CO 1: Compute the coefficient of correlation and regression between two variables, Binomial and Poisson, distributions, Sampling, Level of significance, Chi-square test.

CO 2: Use Laplace transform in solving differential equations

CO 3: Know the concept of analytic functions and its applications in engineering

CO 4: Understand bilinear transformation and its applications

CO 5: Use Cauchy's residue theorem in solving real integrals

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End Term Examination, Even Semester 2021-22

B. Tech. (CS, ME), II Year, IV Semester

Subject Code & Subject Name- BMAS 1103, ENGINEERING MATHEMATICS III

Time: 3 Hours **Maximum Marks: 50**

Instruction for students:

(1) Attempt all the sections.

(2) Marks of the questions and internal choice are indicated in each section.

Section – A

Attempt All Questions. 4 X 5 = 20 Marks

No.	Detail of Question	Marks	CO	BL	KL
1	Find the Laplace transform of $f(t) = \cosh(2t)\sin(4t)$ and hence, evaluate the integral $\int_0^\infty e^{-10t}\cosh(2t)\sin(4t)dt$	4	2	E	C
2	Solve the following initial value problem by the help of Laplace transform: $(D^2 - 2D + 2)\,y = 0,\ y = Dy = 1 \text{ at } t = 0.$	4	2	A	P
3	A skilled typist, on routine work, kept a record of his typing mistakes made per day for 300 days, in following format: Mistakes per day: 0 | 1 | 2 | 3 | 4 | 5 | 6 No. of days: 143 | 90 | 42 | 12 | 9 | 3 | 1 Fit a Poisson distribution to the above data and hence, calculate the theoretical frequencies.	4	1	An	P

4	Prove that the following function is harmonic. $u = x^2 - y^2 - 2xy - 2x + 3y$ Also find the harmonic conjugate of this function.	4	3	U	C
5	Find the residue of the following function at simple poles: $f(z) = \frac{2z^2 + 5}{(z+2)^3(z^2+4)}$, and hence, evaluate the complex line integral $\oint_c f(z)dz$, where, c is the boundary of the square formed with the vertices $1+i, 2+i, 2+2i, 1+2i$. **OR,** The following table gives a classification of a sample of 160 plants of their flower color and flatness of leaves. (4) Test if the flower color is independent of the flatness of leaves. [Given: The value of χ^2 at 5 % level of significance for 1 degree of freedom is 3.841.]	4	4 1	E	F

	Flat leaves	Colored leaves	Total
White flower	99	36	135
Red flower	20	5	25
Total	119	41	160

Section – B

*Attempt **All** Questions.* 3 X 5 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
1	Using Cauchy-Riemann theorem, show that $f(z) = z^3$ is analytic on the whole complex plane.	3	3	R	F
2	Using Milne-Thomson method, find the analytic function $f(z) = u + iv$ if it is given that $v = e^x(x\sin y + y\cos y)$.	3	3	An	C

3	Evaluate the contour integral using Cauchy's residue theorem: $\int_c \frac{2z+1}{z^2+z} dz$, where, c is the circle given by $\mid z \mid = \frac{1}{2}$.	3	5	A	P
4	Expand the following function in Laurent's series valid for $1 < \mid z \mid < 3$: $f(z) = \frac{1}{(z+1)(z+3)}$	3	5	C	M
5	Find the residue of the following function at each of its poles: $f(z) = \frac{1}{(z-1)(z-2)(z-3)}$	3	5	E	P

Section – C

Attempt All Questions 5 X 3 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
1	Evaluate the integral $\int_0^{\pi} \frac{d\theta}{2\cos\theta + 3}$ by method of contour integrals in complex plane.	5	5	A	C
2	Examine the nature of the function $f(z) = \frac{x^3 y(y-ix)}{x^6+y^2}; z \neq 0$ and $f(0) = 0$ Show that $\lim_{z \to 0} \frac{f(z)-f(0)}{z} \to 0$ along radius vector $y = mx$ for any value of m. Is $f(z)$ analytic? Support your answer with proof.	5	3	C	M
3	Evaluate the integral: $\int_0^{1+i} (x - y + i x^2)\, dz$ along (a) the straight line joining z = 0 and z = 1 + i (b) the real axis from z = 0 to z = 1 and then along a line parallel to imaginary axis from z = 1 to z = 1 + i.	5	3	An	P

CO – Course Outcome, BL – Abbreviation for Bloom's Taxonomy Level (R-Remember, U-Understand, A-Apply, An-Analyze, E-Evaluate, C-Create), KL – Abbreviation for Knowledge Level (F-Factual, C-Conceptual, P-Procedural, M-Metacognitive).

Course Outcome

CO1: Understand the basic concepts of the data structure and algorithms.
CO2: Implement of stack using array and linked list and its applications.
CO3: Apply the associated operations in linear data structure like stack, Queue and link list.
CO4: Apply the associated operations in Binary Search Tree, AVL Tree and M- Way Search Tree.
CO5: Understand the basic algorithms such as heap sort, graph traversal, quick sort, AVL trees, and hashing.

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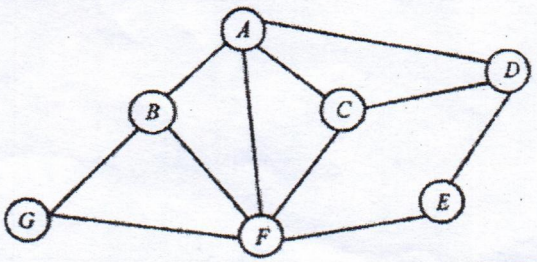
End Term Examination, Even Semester 2021-22
B.Tech (CSE/CCV/DA/IIOT/AIML/CSF), 2nd Year, 4th Semester
BCSC 0006 / Data Structures and Algorithms

Time: 3 Hours **Maximum Marks: 50**

Instruction for students: Write the complete code in one place neatly. Commenting the code is optional.

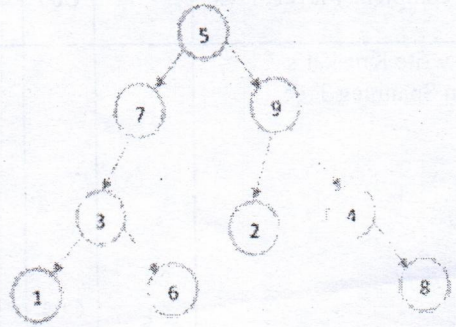
Section – A

Attempt All Questions 4 X 5 = 20 Marks

No.	Detail of Question	Marks	CO	BL	KL
1	Assume that a Singly linked list is implemented with a header node but no tail node. Write a class that includes methods to a) Add a value X if it is not already contained in the linked list. b) Remove the value X if it is contained in the linked list.	2 2	C01 C02	C	C
2	What are the disadvantages of representing a linear queue using array? How are they overcome?	4	CO3	U	C
3	What is a Linked list? What will you prefer to use a singly or a doubly linked lists for traversing through a list of elements?	4	C01 C02	U, An	F
4	Represent following graph in adjacency list. Write the code for breadth first search to traverse the graph. 	4	C04 C05	U, A	C,F
5	Write an algorithm delete a node from AVL Tree.	4	C04	U, C	C

Section – B

*Attempt **All** Questions* 3 X 5 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
1	Explain the various ways in which a graph can be represented bringing out the advantages and disadvantages of each representation.	1.5 1.5	C04	E C	C
2	a) What is the output obtained after preorder, postorder traversal of following tree.  b) Draw the binary tree whose sequential representation is given below.	3	C05	An	M,C

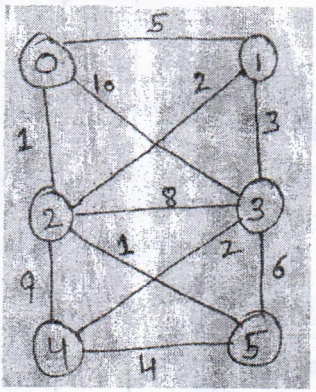
1	2	3	4	5	6	7	8	9	10	11	12	13
A	B	C	D	-	E	F	-	G	-	H	I	-

3	Define Hashing. What is the need for Hashing? Define collision problem in Hashing?	3	C05	C	M,C
4	What is a binary search tree (BST)? Write the code to construct a BST with five nodes.	3	C05 C04	C,A	C
5	Write Short Node on a) AVL Tree b) Threaded Binary Tree c) Minimum Spanning Tree	1 1 1	C05	R	C,F

P.T.O

Section – C

*Attempt **All** Questions* 5 X 3 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
1	Write algorithm for (i) Insertion sort (ii) Bubble sort. Illustrate the insertion sort algorithm and bubble sort algorithm on input [30,20,10,60,80,40]	2.5 2.5	C05	A	CF
2	Why Binary Search algorithm is more efficient that linear search? Depict your answer with suitable example? Mention the time complexity level of two algorithms.	5	C04 C05	U,A	C
3	For the graph given below, write Kruskal's Algorithm to find Minimum Spanning Tree.  	5	C04 C05	A	CF

Course Name: Introduction to Microprocessors

Course Outcome

CO1- Demonstrate the Microprocessor internal architecture and its operations.
CO2- Develop programs based on 8085 microprocessor instruction set and addressing mode.
CO3- Develop program using looping, counting, indexing, counter and time delays.
CO4- Understand the concept of stack and subroutine for modular approach.
CO5- Compare accepted standards and guidelines to select microprocessor (8085 and 8086) to meet performance requirements.
CO6- Analyze the concept of interfacing the processor to external device with I/O programming and Interrupt Driven I/O.
CO7- Understand the working of interfacing chips (8237, 8253/54, 8255 & 8259).

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End Term Examination, Even Semester 2021-22
B.Tech (CSE/AIML/DA/IIOT/CCV/CSF), II-Year, IV-Semester
BCSC 0007: Introduction to Microprocessors

Time: 3 Hours **Maximum Marks: 50**

Instruction for students:

1. All parts of a question should be answer at one place.
2. First complete the one set of section then start a new section.
3. Answer should be brief and to-the-point.
4. Use suitable diagrams to explain the answer.
5. Assume the appropriate data for any missing data.

Section – A

*Attempt **All** Questions* 4 X 5 = 20 Marks

No.	Detail of Question	Marks	CO	BL	KL
1	What is the need of branch group instructions? Explain the unconditional CALL instruction in detail.	4	2	U	C
2	Explain the function of following instructions: (a)LHLD 4000H (b) SPHL (c) OUT port address (d) DAD D	4	2	U	F
3	Write the four differences between 8085 and 8086 microprocessor.	4	5	R	C
4	Explain all the types of Indirect addressing modes of 8086 microprocessor. Also write suitable examples of each addressing mode.	4	2	U	F
5	Explain the concept of memory segmentation in 8086 microprocessor with suitable diagram. What are the advantages of memory segmentation in 8086 microprocessor? (OR) Draw the pin diagram of 8086 microprocessor and show all its pin signals.	4	1	R	P

Section – B

*Attempt **All** Questions* 3 X 5 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
1	Explain all the control flags of 8086 Microprocessor.	3	1	U	F
2	Write an assembly language program for 10H bytes of data are stored in memory locations at 2050H to 205FH. Transfer this entire block of data to new memory location starting at 3050H.	3	3	A	P
3	Draw the internal architecture of 8259. Explain its key features.	3	7	U	C
4	Write an assembly language program to calculate sum of Odd numbers from a series. Length of the series stored in memory location 3000H and the series begins from memory location 3001H. Consider the sum to be 8 bit numbers. Store the sum at memory location 3500H.	3	3	A	P
5	Why microprocessor 8086 is faster than microprocessor 8085? How Pipelining is performed in 8086 microprocessor? Explain with the help of a diagram.	3	5	An	P

Section – C

*Attempt **All** Questions* 5 X 3 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
1	Calculate the delay of given set of instructions. Assume the microprocessor is operating with the frequency of 2 megahertz. MVI B, 20H LOOP2: MVI C, FEH LOOP1: DCR C JNZ LOOP1 DCR B JNZ LOOP2	5	3	E	C
2	Draw the internal architecture of 8255. Mention the control word format of I/O and BSR Mode.	5	7	R	P
3	Explain the following signals of 8086 microprocessor pin diagram. (a) LOCK' (b) TEST' (c) BHE' (d) S0' S1' S2' (e) QS1, QS0	5	1	U	F

Course Name: **Discrete Mathematics**

Course Outcome

CO1: Understand the notion of mathematical thinking and proofs to solve the problem.
CO2: Apply the basics of discrete probability and number theory to solve the real world problem.
CO3: Analyze basic discrete structures and algorithms using effectively algebraic techniques.
CO4: Analyze mathematical concepts like sets, reasoning, relational algebra and graph theory to solve optimization problems.
CO5: Analyze the validity of an argument using logical notation.
CO6: Demonstrate the basic structures of proof techniques to write and evaluate the validity of arguments.
CO7: Understand the basic principles of sets, set equalities and operations in sets.
CO8: Apply counting principles to determine probabilities.

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End Term Examination, Even Semester 2021-22
B.Tech. CSE, II Year, IV Semester
Subject Code- BCSC1010, Subject Name-Discrete Mathematics

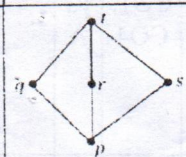
Time: 3 Hours **Maximum Marks: 50**

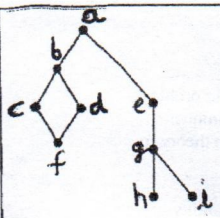
Instruction for students:

1. All parts of a question should be answered at one place.
2. Answer should be brief and to-the-point and be supplemented with neat sketches.
3. Any missing or wrong data may be assumed suitably giving proper justification.

Section – A

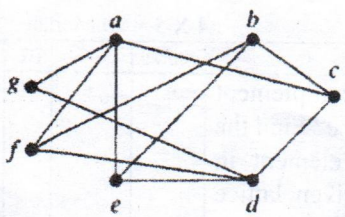
*Attempt **All** Questions* 4 X 5 = 20 Marks

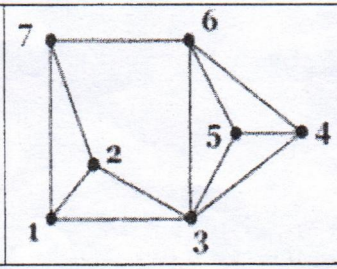
No.	Detail of Question	Marks	CO	BL	KL
1	 What do you mean by complement of an element in a lattice? Find the complement of each element in given lattice. Is the given lattice complemented? OR How many different numbers of six digits (without repetition) can be formed from the digits 3, 1, 7, 0, 9, 5? How many of them will have 0 in the unit place? How many of them are divisible of 5? How many of them are odd numbers?	4	CO1, CO2	U, A	C
2	Consider the following scenario: Suppose the number of bacteria in a colony doubles every hour and a colony begins with five bacteria. To find how many bacteria will be present in n hours, generate a recurrence relation and find its solution.	4	CO1	E	P

3	 In the given Hasse diagram, find the following: (i) Minimal and Maximal elements (ii) Least and Greatest elements (iii) glb and lub of {d,e} and {h,g} (iv) Is it lattice? Justify.	4	CO2	U	M
4	Show that **t** is a valid conclusion from the premises **p→q, q→r, r→s, ~s and p v t**	4	CO6	A	C
5	$<\{1,6\}, X_7>$ is a subgroup of $<Z_7^*, X_7>$. Find the cosets of the subgroup in this group. OR Draw the following graphs: (i) $K_{1,8}$ (ii) W_7 (iii) C_7 (iv) Q_3	4	CO3, CO4	U, A	P

Section – B

*Attempt **All** Questions* 3 X 5 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
1	Define Bipartite graph. Is the given graph bipartite? 	3	CO1, CO4	R, A	C
2	Translate in two ways the following statement into logical expressions using predicates, quantifiers, and logical connectives. First, let the domain consist of the students in your class and second, let it consist of all people. **"There is a student in your class who was not born in California."**	3	CO1, CO4	U, A	M
3	Show that **$(p \rightarrow r) \vee (q \rightarrow r)$ and $(p \wedge q) \rightarrow r$** are logically equivalent.	3	CO1,	R	C
4	Write the contrapositive, converse, and inverse of the conditional statement "**I go to the beach whenever it is a sunny summer day**".	3	CO4	U	C

5	 Find if exists: (i) Euler circuit (ii)Hamiltonian circuit	3	CO4	U	M

Section – C

*Attempt **All** Questions* 5 X 3 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
1	Let G={(a,b) where a,b ϵ R, a ≠ 0} and a binary operation * on G is defined by (a,b)*(c,d)=(ac,bc+d) for all (a.b),(c,d) ϵ G. Determine whether <G,*> is a group?	5	CO5	E	M
2	Find the PDNF and PCNF of **(p^q)v(~p^r)v(q^r)**.	5	CO4	A	P
3	What do you mean by cyclic group? Find the order of each element of abelian group $<Z_5^*, X_5>$. Is $<Z_5^*, X_5>$ cyclic? OR Show that $<Z_5, +_5, X_5>$ is a field.	5	CO5	U	P

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End-Term Examination, Even Semester 2021-22

B.Tech. CSE, II: Year, IV: Semester

BCSE 0901: IIoT Communication & Standard Interface

Time: 1.5-hour **Maximum Marks: 50**

Section A

Note: Attempt all questions. **30 x 1 = 30 Marks**

1. Select the best match for the below function in "wire.h" library?
   wire.read()
   A. Used to read data in I2C communication
   B. Used to read data in SPI communication
   C. Used to read data in UART communication
   D. None of the above

2. Why do we write "**Subscription**" in Thingworx?
   A. To create a new variable
   B. For generating alert
   C. To take some action when an Event occurs
   D. To delete any property

3. How many arguments are passed in "**wire.beginTransmission()**" function using I2C communication?
   A. 1
   B. 2
   C. 3
   D. 0

4. **"wire.available()"** function in "wire.h" library is used for?
   A. To check if data is coming on data line
   B. To ignore the incoming data

C. Both of the above
D. None of the above

5. What does HTTP do?
   A. Enables network resources and reduces perception of latency
   B. Reduces perception of latency and allows multiple concurrency exchange
   C. Allows multiple concurrent exchange and enables network resources
   D. Enables network resources and reduces perception of latency and allows multiple concurrent exchange

6. How many types of message formats are there in HTTP protocol?
   A. Request message
   B. Response message
   C. Both A & B
   D. Only B

7. Which of the following category is used for business to consumer process?
   A. Group IoT
   B. Community IoT
   C. Personal IoT
   D. Industrial IoT

8. Which markup language can be used in restful web API?

   A. JSON AND HTML
   B. JSON and JAVA
   C. HTML and JAVA
   D. JSON and XML

9. "onRequest()" function in wire.h library is?

   A. Executed when a data is sent by Master Device
   B. Executed when a data is requested by Master Device
   C. Both of the above
   D. None of the above

10. "onReceive()" function in wire.h library is?

A. Executed when a data is sent by Master Device
B. Executed when a data is requested by Master Device
C. Both of the above
D. None of the above

11. Select the best match for the below function in "wire.h" library?

wire.read()

A. Used to read data in I2C communication
B. Used to write data in SPI communication
C. Used to read data in UART communication
D. None of the above

12. "WiFiclient.h" library in NodeMCU ESP8266 is used?

A. Access Wi-Fi communication
B. Handle client requests & responses
C. Control data communication over internet
D. All of the above

13. What is default size of data communication in Arduino UNO for I2C communication?

A. 16 byte
B. 32 byte
C. 48 byte
D. 64 byte

14. Why a "style definition" entity is used in Thingworx mashup design ?

A. To enhance the User Interface with required color
B. To delete a state definition
C. To create widgets
D. None of the above

15. "State Definition" Entity is used to?

A. Create Properties in things
B. Write subscription in Thingworx
C. To changes the style definition based on state of data
D. All of the above

16. In Smart Water Management Project, in Thingworx mashup design a **"shape"** widget was used to:

A. Represent the state of the Pump with colour.
B. Turn off the pump
C. Both of the above
D. None of the above

17. **"Valustream"** is used in ThingWorx for:

A. Data security
B. Data Storage and Streaming
C. Data encrypt
D. None of these

18. "Things" does not contain properties:

A. TRUE
B. FALSE

19. In Thingworx "Querypropertyhistory "service is used for:

A. NO-SQL data
B. SQL data
C. Hybrid data
D. None of these

20. "GetProperties" service in Things Enitiy is used to:

A. Modify the property's name
B. Fetch the property's value
C. Both
D. None of above

21. Thing can be created without mapping a project:

A. TRUE
B. FALSE

22. GET method is used to append form data to the URL in name or value pair.

A. TRUE
B. FALSE

23. In the POST method, the data is sent to the server as a package in a separate communication with the processing script.

A. TRUE
B. FALSE

24. PUT method is used to update resource available on the .....:

A. Client
B. Server
C. Both
D. None

25. REST defines a set of constraints for how the architecture of an Internet-scale distributed hypermedia system, such as the Web:

A. TRUE
B. FALSE

26. REST stand for:

A. Representational state transfer
B. Re- state transfer
C. Residual state transfer
D. None of these

27. A URL is a unique sequence of characters that identifies a logical or physical resource used by web technologies.

A. TRUE
B. FALSE

28. The HTTP 205 OK success status response code indicates that the request has succeeded.

A. TRUE
B. FALSE

29. The HyperText Transfer Protocol (HTTP) 400 Bad Request response status code indicates that the server cannot or will not process the request:

A. TRUE
B. FALSE

30. A port number is the ..... address of each application or process that uses a network or the Internet to communicate.

A. Physical
B. Logical
C. Both
D. None of these

Section B

Note: Attempt all questions. **40 x 0.5 = 20 Marks**

31. A server is a type of host that offers resources to the other hosts.

A. FALSE
B. TRUE

32. Mark the best match for difference between a "GenericThingTemplete" and "RemoteThingTemplete".

A. GenericThingTemplete is used for remote devices where as RemoteThingTemplete is used for local device.
B. RemoteThingTemplete is used for remote devices where as GenericThingTemplete is used for local device.
C. None of the above
D. Both of the above

33. When a …….. request is made via a RESTful API, it transfers a representation of the state of the resource to the requester or endpoint.

A. Server
B. Client
C. Both
D. None of these

34. Which one you have used to access Thingworx:

A. http://dcs.glaitm.org:7080/Thingworx
B. https://dcs.glaitm.org:7080/Thingworx
C. http://dcs.glaitm.org:443/Thingworx
D. None of these

35. HTTP Works at Application Layer?

A. True
B. False

36. "wire.h" library allows you to communicate with SPI devices.

A. FALSE
B. TRUE

37. During I2C communication in Arduino to NodeMCU, all the sensor data was concatenated and separated by "," to:

A. Simplify the receiving and separation of data at NodeMCU
B. Because the data size was not constant
C. Both of above
D. None of the above

38. Application Keys, are security tokens used for authentication in ThingWorx.

A. True
B. False

39. In **"wire.begin(address)"** function **"address"** could be passed as:

A. the 7-bit or 10-bit slave address
B. SDA,SCL Pin Number
C. Option A or B
D. None of these

40. In **"wire.requestFrom(address, quantity)"** function, **"quantity"** is:

A. the number of bytes to request.
B. the number of bit to request.
C. the number of words to request.
D. None of these

41. The first line of HTTP request message is called _____

A. Request line
B. Header line
C. Status line
D. Entity line

42. What is the use of the ESP8266 WiFi Module?

A. Monitors Motion
B. Evaluates air pressure
C. Network Provider
D. Switches circuits

43. What is the use of the TX pin?

A. Upload
B. Download
C. Ground
D. Power input

44. What will happen if we supply a voltage of 250 kV to the Vcc of the ESP8266 WiFi Module?

A. Damage is caused
B. Module will shut down
C. Module will not respond for the time the voltage is applied

D. Module will function normally

45. What kind of device is the ESP8266 WiFi Module?

A. Passive Sensor
B. Active Sensor
C. Networking Device
D. Switching Device

46.

```
PUT /new.html HTTP/1.1
Host: example.com
Content-type: text/html
Content-length: 16

<p>New File</p>
```

Is it a proper PUT method example:

A. True
B. False

47. The HTTP GET request method creates a new resource or replaces a representation of the target resource with the request payload.

A. True
B. False

48. To make a Thing in Thingworx, Project Mapping is not mandatory.

A. FALSE
B. TRUE

49. **"getproperties"** Service is used for updating the value of properties.

A. FALSE
B. TRUE

50. **"QueryPropertyHistory"** Service is used to Query No SQL data in Thingworx

A. True
B. False

51. **"Valuestream"** is not mandatory to store historical data.

A. False
B. True

52. **"Logged"** check box belongs in valustream General Information

A. False
B. True

53. To get Timestamp data .....check box should be marked in thing properties:

A. Persistent
B. Read only
C. Logged
D. None of these

54. Thingworx Mashup page offers static page only.

A. False
B. True

55. Data Filter is applied for .....

A. Filter the Noise
B. Select particular Bandwidth of data
C. Eliminate outliers
D. All of these

56. The values GET, POST, HEAD etc. are specified in _____ of HTTP message

A. Request line
B. Header line
C. Status line
D. Entity body

57. "Autorefresh "is a ..... in Thingworx.

A. Tool
B. Widget
C. Both
D. None of these

58. Legends are a useful way to label data series plotted on a graph.

A. False
B. True

59. **"Applicationkey"** are not an entity type.

A. True
B. False

60. To get the historical data ..... Services is required

A. Querypropertyhistory
B. Getproperties
C. Get event properties
D. None of these

61. The _____ method when used in the method field, leaves entity body empty.

A. POST
B. SEND
C. GET
D. PUT

62. Find the oddly matched HTTP status codes

A. 200 OK
B. 400 Bad Request
C. 401 Unauthenticated
D. 304 Not Found

63. URI Stands for:

A. Uniform Resource Images
B. Uniform Resnet Identifier
C. Uniform Resource Identifier
D. None of the above

64. Which of these is a standard interface for serial data transmission?

A. ASCII
B. RS232C
C. 2
D. Centronics

65. NodeMCU ESP8266 is an example of System on chip (SOC).

A. False
B. True

66. Which one is true:

A. WiFi.begin(ssid, password);
B. WiFi.begin(password,ssid);
C. WiFi.begin(9600);
D. None

67. How to get the IP address of client in NodeMCU ESP8266:

A. WiFi.localIP()
B. WiFi.loclIP()
C. WIFI.localIP()
D. None of these

68. How many Master & Slaves can communicate over I2C communication on 7-bit Address mode?

A. 127
B. 256
C. 1023
D. None of the above

69. In Thingworx, ThingTemplete can be designed by user in term of fetching Predefined properties to Things.

A. True
B. False
C. It's possible but with limited properties.
D. All of the above

70. Which of the following is advantage of RESTful web service being stateless?

A. Web services can treat each method request independently.
B. Web services need not to maintain client's previous interactions. It simplifies application design.
C. As HTTP is itself a statelessness protocol, RESTful Web services work seamlessly with HTTP protocol.
D. All of the above.

SET - A

Course Name: English for Professional Purposes II (for, BTech & BTech-Biotech)

Course Outcome

CO1-They will be able to communicate in simple formal situations such as meetings and discussions.
CO2-They will be able to interact in organizational setting with basic level of appropriateness and fluency.
CO3-They will develop a sense of language use in relation to place, person and time.
CO4-They will be able to identify and correct basic level errors in written English.
CO5-They will develop some sense of common errors in English grammar.

Printed Pages: 28 **University Roll No. .....**

End Term Examination, Even Semester 2021-22
BTech & BTech-Biotech, 2nd Year, Semester IV
BELH0004: English for Professional Purposes II

Time: 3 Hours **Maximum Marks: 50**

Instruction for students:

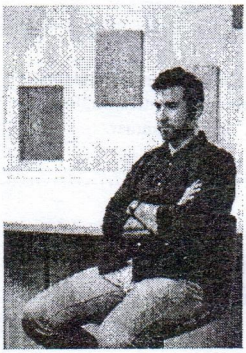
- Answer each of the questions in "**Section A**" on one side of the answer-booklet. That is, start and complete answering each of the questions on a fresh side of your answer booklet.
- Answers to the questions in "Section A" MUST be in the given order of the questions, 1-2-3-4. Otherwise, marks WILL NOT be given.
- All answers must STRICTLY adhere to the instructions given. Or else, marks WILL BE deducted.

Section – A

*Attempt **All** Questions* 4 X 5 = 20 Marks

No.	Detail of Question	Marks	CO	BL	KL
1	Read the following five captions given and choose which of the five images each of the captions most appropriately defines: i. The interviewee looks under-confident due to his posture. ii. The interviewee looks confident with upright posture and nice professional gesture. iii. The interviewee looks confident with appropriate hand gestures during the response. iv. The interviewee seems to be tired due to lack of sufficient rest.	5	1	U	C

v. The interviewee seems to be overconfident and defensive, or disagrees

A. 

B.

C.

	D. E.				
2	Consider yourself as ABC. Now, write a résumé for an internship position that you want to attend in June-July 2022. Invent necessary details. (**Remember**: The résumé must be one full side of your answer booklet. Design it accordingly. You may want to rotate the answer booklet in the portrait mode if you consider so.)	5	4	A	P
3	After writing your résumé, now write an appropriate cover letter to the HR Manager of the company/ institute to which you would apply. (**Remember**: The cover letter must be one full side of your answer booklet. You may want to rotate the answer booklet in the portrait mode if you consider so.)	5	3	A	C
4	Now, considering yourself in front of the interview panel, write ONE PARAGRAPH (approx. 100 words/ half a page), introducing yourself and your strengths. It must be a professional introduction with reference to your résumé and cover letter that you have written above.	5	1	A	C

Section – B

Attempt All Questions 90 X 0.333 = 30 Marks

1. Choose the correct option.
What time ____you _____Mathura Junction?
A. shall, reaches
B. will, reaching
C. shall, be reached
D. are, reach
E. will, reach

2. Choose the correct alternative for the underlined parts of the sentence.
She was reading a book when the light go off.
A. was reading, went off
B. is reading, went off
C. reads, went off
D. reads, going off
E. no correction required

3. Choose the correct alternative for the underlined parts of the sentence.
I will have my secretary to sent you an email regarding the placement details.
A. will have, send
B. would have, sends
C. will have, is sending
D. would have, sending
E. no correction required

4. Choose the correct change of narration from the options:
Payal said to Monika, "She is better than she was".
A. Payal told Monika that she is better than she had been.
B. Payal told Monika that she was better than she was.

C. Payal told Monika that she was better than she has been.
D. Payal told Monika that she was better than she had been.
E. Payal told Monika that she is better than she was.

5. Choose the correct change of narration from the options:
I said to my servant, "At what time did you go to the station yesterday?"

A. I asked my servant at what time he went to the station the previous day.
B. I asked my servant at what time he had gone to the station yesterday.
C. I asked my servant at what time he had gone to the station the previous day.
D. I asked my servant at what time he had gone to the station that day.
E. I asked my servant at what time he had gone to the station the next day.

6. Choose the most logical order of following sentences:
A. Since then, intelligence tests have been mostly used to separate dull children in school from average or bright children, so that special education can be provided to the dull.
B. In other words, intelligence tests give us a norm for each age.
C. Intelligence is expressed as intelligence quotient, and tests are developed to indicate what an average child of a certain age can do: what a 5-year-old can answer, but a 4-year-old cannot, for instance.
D. Binet developed the first set of such tests in the early 1900s to find out which children in school needed special attention.
E. Intelligence can be measured by tests.

A. CDABE
B. DECAB
C. ECDAB
D. CBADE
E. ABCDE

7. Select the correct response for the sentence: "Being open to the sun, the windowsill is the proper place for an indoor man's plants."

A. This sentence contains a dangling modifier. The correct subject need to be inserted after this dangling modifier to correct it.

B. There is no error and the sentence should remain unchanged.

C. The sentence contains a misplaced modifier. Consider reorganizing the phrase: "indoor man's plants".

D. This sentence contains squinting modifier.

E. None of the responses is correct.

8. Choose the correct alternative for the underlined parts of the sentence.

Did your boss been have you worked late last night?

A. had, works

B. has, worked

C. have, is working

D. have, work

E. no correction required

9. Locate which part is erroneous, if any:

She wore (A)/ a bicycle helmet (B)/ on her head (C)/ that was too large. (D)/ No Error (E)

10. Choose the correct option:

If he _____ hard, he could have earned enough money.

A. work

B. worked

C. have worked

D. had worked

E. had been worked

11. Choose the correct answer:

A. If the child is hungry, it cries.

B. If the child was hungry, it will cry.

C. If the child had been hungry, it would have cried.

D. A and C are correct.

E. Only A is correct.

12. Choose the correct option:

If it _____, Nina _____ an umbrella with her.

A. rain/take

B. rains/would take

C. rained/would take

D. rained/took

E. rained/would have taken

13. Locate which part is erroneous, if any:

Not expecting rain (A)/, the umbrella (B)/ was propped (C)/ up against (D)/ the wall (E).

14. Which option most logically completes the sentence? "Having driven the car around the block, _____"

A. notebook crashed.

B. Rajan crashed it.

C. the feather crashed it.

D. the tree began to sway.

E. None of the options is correct.

15. Choose the most logical order of following sentences:

A. He was so busy with them that he did not get time to eat.
B. Thousands of people came to him and asked different types of questions.
C. No one cared to see that he had his food or rest that night.
D. Swami Vivekananda once stayed in a small village.

A. BCDA
B. CBAD
C. DBAC
D. CADB
E. ABCD

16. Choose the correct change of narration from the options:

The teacher said to the students, "Do you remember the lesson well?"
A. The teacher asked the students if they remember the lesson well.
B. The teacher asked the students if he remembered the lesson well.
C. The teacher asked the students if they remembered the lesson well.
D. The teacher asked the students if I remembered the lesson well.
E. The teacher asked the students if I remember the lesson well.

17. Select the appropriate brief expression for the following:

Agreeing on something which is being done or suggested.
A. on the same page
B. on the same line
C. agree
D. sharing same thought
E. optimistic

18. Choose the correct alternative for the underlined parts of the sentence.
My teacher made me read five novels in twenty days.
A. make, reads
B. has make, read
C. making, is reading
D. make, reading
E. no correction required

19. Choose the correct alternative for the underlined part of the sentence.
After walking a few steps in the forest, our guide sensing some sort of movement in the foliage behind us.
A. sensed
B. is sensing
C. had a sense
D. was sensed
E. no correction required

20. Choose the correct option.
I_____ as soon as I complete my work.
A. leave
B. shall leave
C. shall been leaving
D. going to leave
E. will be left

21. Choose which part of the sentence(s) has an erroneous usage of tense. If correct, mark as "No correction required."
If you have heard of Cryptocurrency, (A)/ you kñowing how Bitcoin and other altcoins are becoming popular among student investors. (B)/ Knowing the function and value of these digital currencies is vital (C)/ before someone dabbles in them. (D)/ No correction required. (E)

22. Choose the correct option:
If you freeze water, it _____ solid.
A. become
B. becomes
C. will become
D. became
E. had become

23. Locate which part is erroneous, if any:
Let's (A)/ adopt (B)/ the dog from (C)/ the animal shelter (D)/ with black spots (E).

24. Which of the following sentences is correct?
A. Having jumped up, the cat pounced.
B. The cat pounced when the mouse jumped up.
C. Having jumping up, the cat pounced the mouse.
D. Jumping up the cat, the mouse pounced.
E. None of the options is correct.

25. Choose the most logical order of following sentences:
A. Both parties use capital and labour in the struggle to secure property rights.
B. The thief spends time and money in his attempt to steal (he buys wire cutters) and the legitimate property owner expends resources to prevent the theft (he buys locks).
C. A social cost of theft is that both the thief and the potential victim use resources to gain or maintain control over property.
D. These costs may escalate as a type of technological arms race unfolds.
E. A bank may purchase more and more complicated and sophisticated safes, forcing safe-crackers to invest further in safe-cracking equipment.
A. ABCDE
B. CABED
C. ACBED
D. CBEDA
E. EDCBA

26. Choose the correct change of narration from the options:
He said to me, "Honesty is the best policy."
A. He told me that honesty was the best policy.
B. He told me that honesty should be the best policy.
C. He tells me that honesty is the best policy.
D. He told me that honesty is the best policy.
E. He says that honesty is the best policy.

27. Select the appropriate brief expression for the following:
Tasks which are easily achieved.
A. easy job
B. low-hanging fruit
C. mean job
D. handsome job
E. probability of happening

28. Choose the correct alternative for the underlined parts of the sentence.
His reaction did forced us to rethinking our policy.
A. forced, to rethink
B. forcing, to rethinked
C. force, to rethought
D. had been forced, think
E. no correction required

29. Choose the correct alternative for the underlined part of the sentence.
When he came in, I write a letter.
A. writing
B. was writing
C. is writing
D. writes
E. no correction required

30. Choose the appropriate answer:
Rafael Nadal _____ his tie with his greatest rivals, Roger Federer and Novak Djokovic.

A. break
B. broke
C. broken
D. have broken
E. has been breaking

31. Choose the correct option:
I have no fire insurance. If my house _____ down, I would have no way to buy new furniture.

A. burns
B. burned
C. had burned
D. has burned
E. None of these

32. Locate which part is erroneous, if any:
The salesman (A)/ tried (B)/ to sell (C)/ a car to a customer (D)/ that was broken down (E).

33. In the sentence that follows, a section has been underlined. You are required to replace it with the correct idiom.
Examining job applications is the first stage of examining a group of people and decide who are suitable and who are not.

A. examining the cats and dogs
B. examining the real problem
C. examining the bulls and bears
D. separating the cats and dogs
E. separating the sheep from the goats

34. Choose the most logical order of following sentences:

A. It also gives rise to a feeling of animosity among the different sections of the society.

B. In a democratic system, frequent use of power is never desirable, be it on the part of government or the people.

C. Therefore, citizens should never resort to violent ways and means in democracy, though they have the right to oppose the government.

D. It destroys the stability and security in public life.

A. DBAC
B. BDCA
C. BDAC
D. DACB
E. DCBA

35. Choose the correct change of narration from the options:

Abhishek said to him, "Your team will win you the first prize."

A. Abhishek told him that his team would win him the first prize.
B. Abhishek told him that his team will win him the first prize.
C. Abhishek told him that your team would win you the first prize.
D. Abhishek told him that my team would win me the first prize.
E. Abhishek told him that your team will win you the first prize.

36. Select the appropriate brief expression for the following:

Examining something in great details.

A. introspection
B. fecundity
C. drill down
D. plumage
E. perfectionist

37. Choose the correct alternative for the underlined parts of the sentence.

I don't let my kids watching violent movies.
A. let, watched
B. letting, watch
C. let, is watching
D. let, watch
E. no correction required

38. Choose the correct alternative for the underlined part of the sentence.

You learn English for six years now.
A. has learnt
B. learns
C. have learning
D. have been learning
E. no correction required

39. Choose the correct option:

If he _____ thirsty, he would have drunk some water.
A. was
B. had been
C. is
D. were
E. has been

40. Choose the correct option:

I _____ happy if I _____ more free time.
A. will/have
B. was/would be
C. would be/was
D. would be/had
E. had been/would have

41. Locate which part is erroneous, if any:

He stood up (A)/ on the pedals (B)/ of his bike to get extra power (C)/ as he cycled up the hill. (D)/ No Error (E)

42. Choose the most logical order of following sentences:

A. In the case of King Merolchazzar's courtship of the Princess of the Outer Isles, there occurs a regrettable hitch.
B. She acknowledges the gifts, but no word of a meeting date follows.
C. The monarch, hearing good reports of a neighboring princess, dispatches messengers with gifts to her court, beseeching an interview.
D. The princess names a date, and a formal meeting takes place; after that everything buzzes along pretty smoothly.
E. Royal love affairs in olden days were conducted on the correspondence method.

A. ACBDE
B. ABCDE
C. ECDAB
D. ECBAD
E. CAEDB

43. Choose the correct change of narration from the options:

My friend said to me, "Do not sit idle and do not waste your time".
A. My friend invited me not to sit idle and not to waste my time.
B. My friend said that me not to sit idle and not to waste your time.
C. My friend advised me not to sit idle and not to waste my time.
D. My friend eulogised me not to sit idle and not to waste my time.
E. My friend ordered me not to sit idle and not to waste your time.

44. Choose the correct change of narration from the options:
Nakul said to me, "Your project may be very effective but you will not be able to convince Mr. Nikhil Agrawal."
A. Nakul told me that your project may be very effective but you will not be able to convince Mr. Nikhil Agrawal.
B. Nakul told me that my project might be very effective but I will not be able to convince Mr. Nikhil Agrawal.
C. Nakul told me that your project may be very effective but you would not be able to convince Mr. Nikhil Agrawal.
D. Nakul told me that my project may be very effective but I would not be able to convince Mr. Nikhil Agrawal.
E. Nakul told me that my project might be very effective but I would not be able to convince Mr. Nikhil Agrawal.

45. Find out the idiom for the following:
Someone who is good at calculating numbers.
A. number expert
B. number cruncher
C. accountant
D. mathematician
E. numismatic

46. Choose the correct option.
I _____the electrician_____at my broken light.
A. having, look
B. had, been looked
C. had, look
D. have, is looking
E. has, look

47. Choose the correct option.
As soon as the old clock on the town hall _____ midnight, it _____quiet in the street.

A. struck, becames
B. is struck, became
C. was struck, becoming
D. has strucks, become
E. struck, became

48. Choose the correct option:
You will catch the train if you _____ earlier.
A. left
B. leaves
C. leave
D. would leave
E. had left

49. Locate which part is erroneous, if any:
The pedestrian (A)/ was hit (B)/ by a car (C)/, sitting on the curb. (D)/ No Error (E)

50. Locate which part is erroneous, if any:
He struck (A)/ the fishbowl (B)/ with his forehead, (C)/ which fortunately was empty. (D)/ No error.

51. Choose the most logical order of following sentences:
A. To much of the Labour movement, it symbolizes the brutality of the upper classes.
B. And to everybody watching, the current mess over foxhunting symbolizes the government's weakness.
C. To foxhunting's supporters, Labour's 1991 manifesto commitment to ban it symbolizes the party's metropolitan roots and hostility to the countryside.
D. Small issues sometimes have large symbolic power.
E. To those who enjoy thundering across the countryside in red coats after foxes, foxhunting symbolises the ancient roots of rural lives.

A. DEACB
B. ECDBA
C. CEADB
D. DBAEC
E. BEADC

52. Choose the correct change of narration from the options:
I said to Rajat, "Let us play football."
A. I suggested Rajat that let us play football.
B. I suggested Rajat that I should play football.
C. I suggested Rajat that let me play football.
D. I suggested Rajat that he should play football.
E. I suggested Rajat that we should play football.

53. Choose the correct change of narration from the options:
She says, "I study everyday in the morning."
A. She says that she studies everyday in the morning.
B. She says that she studied everyday in the morning.
C. She says that she study everyday in the morning.
D. She will say that she studies everyday in the morning
E. She says that I study everyday in the morning.

54. Find out the idiom for the following:
An employee is paid the necessary minimum salary.
A. paid peanuts
B. less wages employee
C. underpaid employee
D. paid nuts
E. bonded labour

55. Choose the correct option.
My mother never _____us _____ food with dirty hands.

A. let, eaten
B. let, eat
C. letting, eats
D. let, to ate
E. let, eating

56. Choose the correct option.
If it _____, we shall not go out.
A. raining
B. will rain
C. rains
D. will be rain
E. is raining

57. Choose the correct option:
If they had not _____ the car, I would have driven you.
A. take
B. taken
C. would take
D. took
E. takes

58. Locate which part is erroneous, if any:
Raj fell (A)/ asleep (B)/ under a tree (C), tired from the long hike up the mountain. (D)/ No Error (E)

59. Which of the options is correct about the given sentences?
Sentence 1: I took my car to the mechanic with the bad tires.
Sentence 2: Running down the track, many spectators could be heard.

A. Sentence 1 has erroneous use of modifier.
B. Sentence 2 has erroneous use of modifier.
C. Both the sentences have erroneous use of modifiers.
D. Sentence 1 has a dangling modifier while in Sentence 2 the modifier is misplaced.
E. The sentences are correct.

60. Choose the most logical order of following sentences:
A. But this does not mean that death was the Egyptians' only preoccupation.
B. Even papyri come mainly from pyramids.
C. Most of our traditional sources of information about the Old Kingdom are monuments of the rich like pyramids and tombs.
D. Houses in which ordinary Egyptians lived have not been preserved, and when most people died they were buried in simple graves.
E. We know infinitely more about the wealthy people of Egypt than we do about the ordinary people, as most monuments were made for the rich.
A. CDBEA
B. ECDAB
C. EDCBA
D. DECAB
E. BADCE

61. Choose the correct change of narration from the options:
Mohit asked his father, "Where else can we go?"
A. Mohit asked his father where else they could go.
B. Mohit asked his father where else we could go.
C. Mohit asked his father where else they can go.
D. Mohit asked his father where else we can go.
E. Mohit asked his father where else he can go.

62. Choose the correct change of narration from the options:

Our teacher said to me, "This is your mistake."
A. Our teacher told me that this was your mistake.
B. Our teacher told me that that was my mistake.
C. Our teacher told me that was your mistake.
D. Our teacher told me that that is my mistake.
E. Our teacher told me that this was my mistake.

63. Find out the idiom for the following:
A distorted information or rumour/ gossip about something.
A. Chinese proverb
B. rumour
C. gossip
D. Chinese whispers
E. falsehood

64. Choose the correct option.
He _____me _____my project in due time.
A. has help, wrote
B. help, write
C. helps, written
D. helping, to wrote
E. helped, write

65. Choose the correct option.
Our professor _____to Cambridge last year.
A. visited at
B. went
C. had been gone
D. gone
E. has been gone

66. Choose the correct option:
What will you do if you _____ the history exam?
A. would fail
B. will fail
C. fail
D. failed
E. fails

67. Locate which part is erroneous, if any:
Subjected to humiliation, (A)/ the producers of the reality television show (B)/ should have anticipated (C)/ the contestant might break the contract. (D)/ No Error (E)

68. Which sentence has an example of misplaced modifier?
A. At the lake, Ajay caught a fish that was very small.
B. To win the race, much determination is needed.
C. Yesterday she promised to call me.
D. Karl Benz has, by most automobile historians, been given credit for the invention of the automobile.
E. None of the above sentences has a misplaced modifier.

69. Choose the most logical order of following sentences:
A. Luckily the tide of battle moved elsewhere after the American victory at Midway and an Australian victory over Japan at Milne Bay.
B. It could have been no more than a delaying tactic.
C. The Australian military, knowing the position was hopeless, planned to fall back to the south-east in the hope of defending the main cities.
D. They had captured most of the Soloman Islands and much of New Guinea, and seemed poised for an invasion.
E. Not many people outside Australia realize how close the Japanese got.
A. EDCBA
B. ECDAB
C. ADCBE

D. CDBAE
E. ABCDE

70. Choose the correct change of narration from the options:
He said to me, "Why have you come here?"
A. He asked me why I had come here.
B. He asked me why I had gone there.
C. He asked me why I had come there.
D. He asked me if I had gone here.
E. He asked me whether I had come here.

71. Choose the correct change of narration from the options:
Priyanka says, "I do not work as a freelancer."
A. Priyanka says that she does not work as a freelancer.
B. Priyanka says that she did not work as a freelancer.
C. Priyanka said that she did not worked as a freelancer.
D. Priyanka said that she does not work as a freelancer.
E. Priyanka says that she never worked as a freelancer.

72. Find out the idiom for the following:
Learn to do something with expertise.
A. learn the details
B. knows the job
C. learn the ropes
D. expert in field
E. mastery

73. Choose the correct option.
Ram _____his appointment _____next week.
A. got, schedule
B. get, has scheduled
C. get, is scheduling
D. got, scheduled
E. gotten, schedule

74. Choose the correct option.
If you_____ him, he _____helped you.
A. request, will be
B. requesting , would
C. have request, would have
D. requested, will has
E. had requested, would have

75. Choose the correct option:
If I eat peanut butter, I _____ sick.
A. would have gotten
B. would get
C. get
D. got
E. gets

76. Locate which part is erroneous, if any:
The no-duty rule(A)/ should not apply (B)/ to a patron struck by a ball (C)/ batted by a player during the seventh inning stretch (D)/ while seated in an open concourse. (E)

77. Choose the sentence that does NOT include a dangling modifier.
A. Being a sit-down comedian, it was hard to make ends meet.

B. Biking to class, the sun shined upon the student.
C. Sitting on the threshold, a snake bit him.
D. Devouring ample amounts of cookout, the engineer felt joy seeping back into his soul.
E. All the above sentences have dangling modifiers.

78. Choose the most logical order of following sentences:
A. This feeling of an extensive group gives rise to a fellow feeling, a feeling of brotherhood amongst the citizens.
B. This feeling takes up beyond the bounds of family, caste, religion as well as region and helps us develop a broad perspective that all of us together constitute an extensive group called 'the nation'.
C. National integration is the feeling among all the citizens of a country that they all are part of one nation.
D. We do not, then, limit our thinking to our own caste or religion but think about all our fellow citizens.
A. CDAB
B. CABD
C. CDBA
D. CBDA
E. CBAD

79. Choose the correct change of narration from the options:
She said to me, "Let us visit the market".
A. She suggested me that they should visit the market.
B. She suggested me that I should visit the market.
C. She suggested me that we should visit the market.
D. She suggested me that we would visit the market.
E. She suggested me that we shall visit the market.

80. Choose the correct change of narration from the options:
I said to my mother, "I will take only a cup of tea because I am not hungry".
A. I told my mother that I will take a cup of tea because I was not hungry.
B. I told my mother that I would take a cup of tea because I am not hungry.
C. I told my mother that I would take a cup of tea because I was not hungry.
D. I told my mother that I will take a cup of tea because I am not hungry.
E. I told my mother that I would have taken a cup of tea because I was not hungry.

81. Find out the idiom for the following:
A dismissal or end of contract paper.
A. grievance letter
B. walking letter
C. dismissal paper
D. walking paper
E. walk over paper

82. Choose the correct option.
The hijacker _____the pilots _____the plane in a different direction.
A. force, to took
B. forcing, taking
C. forced, to take
D. forced, took
E. force, is taking

83. Choose the correct option.
Ram _____ in Mumbai for 15 years.
A. is lived
B. has living
C. is been living
D. has been living
E. had living

84. Choose the correct option:
What would you do if it _____ on your wedding day?
A. rained
B. will rain
C. would rain
D. rains
E. had rained

85. Locate which part is erroneous, if any:
The patient was (A)/ referred (B)/ to the physician (C)/ with stomach pains. (D)/ No Error (E)

86. Rohan wants to tell Reena that he loves her and to no one else. What sentence should Rohan use to convey the correct meaning to Reena? Careful attention should be given to the modifier "just".
A. "I have chosen to love just you."
B. "I love you just."
C. "I have just chosen to love you."
D. "I just love you."
E. None of the options is correct.

87. Choose the most logical order of following sentences:
A. The peasant, the shoemaker, the sweeper and such other lower classes of India have much greater capacity for work and self-reliance than you.
B. Remember that the nation lives in the cottage.
C. They are producing the entire wealth of the land without a word of complaint.
D. This process of production is going through long ages.
A. BDAC
B. BDCA
C. DCBA
D. BACD
E. ADCB

88. Choose the correct change of narration from the options:
We said to him, "Please let us do our work".
A. We requested him to let us do our work.
B. We requested him to let me do my work.
C. We requested him that we should do our work.
D. We requested him to let him do his work.
E. We requested him to let them do their work.

89. Choose the correct option.
Prabir had his mobile_____.
A. has repaired
B. let repair
C. repaired
D. got repairing
E. repairs

90. Choose the correct change of narration from the options:
My mother says, "I do not laugh at Neha. She is my good friend".
A. My mother tells that she does not laugh at Neha. She is her good friend.
B. My mother tells that she did not laugh at Neha. She was my good friend.
C. My mother tells that she had not laughed at Neha. She was my good friend.
D. My mother tells that she did not laught at Neha. She is my good friend.
E. My mother tells that she does not laught at Neha. She was my good friend.

SET - B

Course Name: English for Professional Purposes II (for, BTech & BTech-Biotech)

Course Outcome

CO1-They will be able to communicate in simple formal situations such as meetings and discussions.
CO2-They will be able to interact in organizational setting with basic level of appropriateness and fluency.
CO3-They will develop a sense of language use in relation to place, person and time.
CO4-They will be able to identify and correct basic level errors in written English.
CO5-They will develop some sense of common errors in English grammar.

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End Term Examination, Even Semester 2021-22
BTech & BTech-Biotech, 2nd Year, Semester IV
BELH0004: English for Professional Purposes II

Time: 3 Hours **Maximum Marks: 50**

Instruction for students:

- Answer each of the questions in "**Section A**" on one side of the answer-booklet. That is, start and complete answering each of the questions on a fresh side of your answer booklet.
- Answers to the questions in "Section A" MUST be in the given order of the questions, 1-2-3-4. Otherwise, marks WILL NOT be given.
- All answers must STRICTLY adhere to the instructions given. Or else, marks WILL BE deducted.

Section – A

*Attempt **All** Questions* 4 X 5 = 20 Marks

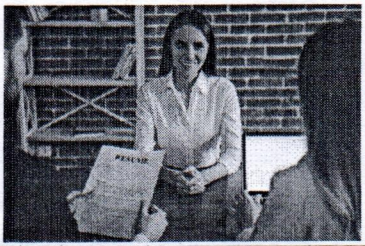
No.	Detail of Question	Marks	CO	BL	KL
1	Read the following five captions given and choose which of the five images each of the captions most appropriately defines: i. The interviewee looks under-confident due to his posture. ii. The interviewee looks confident with upright posture and nice professional gesture. iii. The interviewee looks confident with appropriate hand gestures during the response. iv. The interviewee seems to be tired due to lack of sufficient rest.	5	1	U	C

v. The interviewee seems to be overconfident and defensive, or disagrees

A.

B.

C.

	D. E. 				
2	Consider yourself as ABC. Now, write a résumé for an internship position that you want to attend in June-July 2022. Invent necessary details. (**Remember**: The résumé must be one full side of your answer booklet. Design it accordingly. You may want to rotate the answer booklet in the portrait mode if you consider so.)	5	4	A	P
3	After writing your résumé, now write an appropriate cover letter to the HR Manager of the company/ institute to which you would apply. (**Remember**: The cover letter must be one full side of your answer booklet. You may want to rotate the answer booklet in the portrait mode if you consider so.)	5	3	A	C
4	Now, considering yourself in front of the interview panel, write ONE PARAGRAPH (approx. 100 words/ half a page), introducing yourself and your strengths. It must be a professional introduction with reference to your résumé and cover letter that you have written above.	5	1	A	C

Section – B

Attempt All Questions 90 X 0.333 = 30 Marks

1. Choose the correct option.
If it _____, we shall not go out.
A. raining
B. will rain
C. rains
D. will be rain
E. is raining

2. Choose the correct option:
If they had not _____ the car, I would have driven you.
A. take
B. taken
C. would take
D. took
E. takes

3. Locate which part is erroneous, if any:
Raj fell (A)/ asleep (B)/ under a tree (C), tired from the long hike up the mountain. (D)/ No Error (E)

4. Which of the options is correct about the given sentences?
Sentence 1: I took my car to the mechanic with the bad tires.
Sentence 2: Running down the track, many spectators could be heard.

A. Sentence 1 has erroneous use of modifier.
B. Sentence 2 has erroneous use of modifier.
C. Both the sentences have erroneous use of modifiers.
D. Sentence 1 has a dangling modifier while in Sentence 2 the modifier is misplaced.
E. The sentences are correct.

5. Choose the most logical order of following sentences:

A. But this does not mean that death was the Egyptians' only preoccupation.
B. Even papyri come mainly from pyramids.
C. Most of our traditional sources of information about the Old Kingdom are monuments of the rich like pyramids and tombs.
D. Houses in which ordinary Egyptians lived have not been preserved, and when most people died they were buried in simple graves.
E. We know infinitely more about the wealthy people of Egypt than we do about the ordinary people, as most monuments were made for the rich.

A. CDBEA
B. ECDAB
C. EDCBA
D. DECAB
E. BADCE

6. Choose the correct change of narration from the options:
Mohit asked his father, "Where else can we go?"
A. Mohit asked his father where else they could go.
B. Mohit asked his father where else we could go.
C. Mohit asked his father where else they can go.
D. Mohit asked his father where else we can go.
E. Mohit asked his father where else he can go.

7. Locate which part is erroneous, if any:
She wore (A)/ a bicycle helmet (B)/ on her head (C)/ that was too large. (D)/ No Error (E)

8. Choose the correct option:
If he _____ hard, he could have earned enough money.
A. work
B. worked
C. have worked
D. had worked
E. had been worked

9. Choose the correct answer:
A. If the child is hungry, it cries.
B. If the child was hungry, it will cry.
C. If the child had been hungry, it would have cried.
D. A and C are correct.
E. Only A is correct.

10. Choose the correct option:
If it _____, Nina _____ an umbrella with her.
A. rain/take
B. rains/would take
C. rained/would take
D. rained/took
E. rained/would have taken

11. Locate which part is erroneous, if any:
Not expecting rain (A)/, the umbrella (B)/ was propped (C)/ up against (D)/ the wall (E).

12. Which option most logically completes the sentence? "Having driven the car around the block, _____"
A. notebook crashed.
B. Rajan crashed it.
C. the feather crashed it.
D. the tree began to sway.
E. None of the options is correct.

13. Choose the most logical order of following sentences:
A. He was so busy with them that he did not get time to eat.
B. Thousands of people came to him and asked different types of questions.
C. No one cared to see that he had his food or rest that night.
D. Swami Vivekananda once stayed in a small village.
A. BCDA
B. CBAD
C. DBAC
D. CADB

E. ABCD

14. Choose the correct change of narration from the options:
The teacher said to the students, "Do you remember the lesson well?"
A. The teacher asked the students if they remember the lesson well.
B. The teacher asked the students if he remembered the lesson well.
C. The teacher asked the students if they remembered the lesson well.
D. The teacher asked the students if I remembered the lesson well.
E. The teacher asked the students if I remember the lesson well.

15. Find out the idiom for the following:
A dismissal or end of contract paper.
A. grievance letter
B. walking letter
C. dismissal paper
D. walking paper
E. walk over paper

16. Choose the correct option.
The hijacker _____the pilots _____the plane in a different direction.
A. force, to took
B. forcing, taking
C. forced, to take
D. forced, took
E. force, is taking

17. Choose the correct option.
Ram _____ in Mumbai for 15 years.
A. is lived
B. has living
C. is been living
D. has been living
E. had living

18. Choose the correct option:
What would you do if it _____ on your wedding day?
A. rained
B. will rain
C. would rain
D. rains
E. had rained

19. Locate which part is erroneous, if any:
The patient was (A)/ referred (B)/ to the physician (C)/ with stomach pains. (D)/ No Error (E)

20. Rohan wants to tell Reena that he loves her and to no one else. What sentence should Rohan use to convey the correct meaning to Reena? Careful attention should be given to the modifier "just".
A. "I have chosen to love just you."
B. "I love you just."
C. "I have just chosen to love you."
D. "I just love you."
E. None of the options is correct.

21. Choose the most logical order of following sentences:
A. The peasant, the shoemaker, the sweeper and such other lower classes of India have much greater capacity for work and self-reliance than you.
B. Remember that the nation lives in the cottage.
C. They are producing the entire wealth of the land without a word of complaint.
D. This process of production is going through long ages.
A. BDAC
B. BDCA
C. DCBA
D. BACD
E. ADCB

22. Choose the correct change of narration from the options:
We said to him, "Please let us do our work".

A. We requested him to let us do our work.
B. We requested him to let me do my work.
C. We requested him that we should do our work.
D. We requested him to let him do his work.
E. We requested him to let them do their work.

23. Choose the correct option.
Prabir had his mobile_____.
A. has repaired
B. let repair
C. repaired
D. got repairing
E. repairs

24. Choose the correct change of narration from the options:
My mother says, "I do not laugh at Neha. She is my good friend".
A. My mother tells that she does not laugh at Neha. She is her good friend.
B. My mother tells that she did not laugh at Neha. She was my good friend.
C. My mother tells that she had not laughed at Neha. She was my good friend.
D. My mother tells that she did not laught at Neha. She is my good friend.
E. My mother tells that she does not laught at Neha. She was my good friend.

25. Choose the correct option.
What time ____you _____Mathura Junction?
A. shall, reaches
B. will, reaching
C. shall, be reached
D. are, reach
E. will, reach

26. Choose the correct alternative for the underlined parts of the sentence.
She was reading a book when the light go off.
A. was reading, went off
B. is reading, went off
C. reads, went off
D. reads, going off
E. no correction required

27. Choose the correct alternative for the underlined part of the sentence.
After walking a few steps in the forest, our guide sensing some sort of movement in the foliage behind us.
A. sensed
B. is sensing
C. had a sense
D. was sensed
E. no correction required

28. Choose the correct option.
I_____ as soon as I complete my work.
A. leave
B. shall leave
C. shall been leaving
D. going to leave
E. will be left

29. Choose which part of the sentence(s) has an erroneous usage of tense. If correct, mark as "No correction required."

If you have heard of Cryptocurrency, (A)/ you knowing how Bitcoin and other altcoins are becoming popular among student investors. (B)/ Knowing the function and value of these digital currencies is vital (C)/ before someone dabbles in them. (D)/ No correction required. (E)

30. Choose the correct option:
If you freeze water, it _____ solid.
A. become
B. becomes
C. will become
D. became
E. had become

31. Locate which part is erroneous, if any:
Let's (A)/ adopt (B)/ the dog from (C)/ the animal shelter (D)/ with black spots (E).

32. Which of the following sentences is correct?
A. Having jumped up, the cat pounced.
B. The cat pounced when the mouse jumped up.
C. Having jumping up, the cat pounced the mouse.
D. Jumping up the cat, the mouse pounced.
E. None of the options is correct.

33. Choose the most logical order of following sentences:
A. Both parties use capital and labour in the struggle to secure property rights.
B. The thief spends time and money in his attempt to steal (he buys wire cutters) and the legitimate property owner expends resources to prevent the theft (he buys locks).
C. A social cost of theft is that both the thief and the potential victim use resources to gain or maintain control over property.
D. These costs may escalate as a type of technological arms race unfolds.
E. A bank may purchase more and more complicated and sophisticated safes, forcing safe-crackers to invest further in safe-cracking equipment.
A. ABCDE
B. CABED
C. ACBED
D. CBEDA
E. EDCBA

34. Select the appropriate brief expression for the following:
Examining something in great details.
A. introspection
B. fecundity
C. drill down
D. plumage
E. perfectionist

35. Choose the correct alternative for the underlined parts of the sentence.
I don't <u>let</u> my kids <u>watching</u> violent movies.
A. let, watched
B. letting, watch
C. let, is watching
D. let, watch
E. no correction required

36. Choose the correct alternative for the underlined part of the sentence.
You <u>learn</u> English for six years now.
A. has learnt
B. learns
C. have learning
D. have been learning
E. no correction required

37. Choose the correct option:
If he _____ thirsty, he would have drunk some water.
A. was
B. had been
C. is
D. were
E. has been

38. Choose the correct option:

I _____ happy if I _____ more free time.

A. will/have

B. was/would be

C. would be/was

D. would be/had

E. had been/would have

39. Locate which part is erroneous, if any:

He stood up (A)/ on the pedals (B)/ of his bike to get extra power (C)/ as he cycled up the hill. (D)/ No Error (E)

40. Choose the most logical order of following sentences:

A. In the case of King Merolchazzar's courtship of the Princess of the Outer Isles, there occurs a regrettable hitch.

B. She acknowledges the gifts, but no word of a meeting date follows.

C. The monarch, hearing good reports of a neighboring princess, dispatches messengers with gifts to her court, beseeching an interview.

D. The princess names a date, and a formal meeting takes place; after that everything buzzes along pretty smoothly.

E. Royal love affairs in olden days were conducted on the correspondence method.

A. ACBDE

B. ABCDE

C. ECDAB

D. ECBAD

E. CAEDB

41. Choose the correct change of narration from the options:

My friend said to me, "Do not sit idle and do not waste your time".

A. My friend invited me not to sit idle and not to waste my time.

B. My friend said that me not to sit idle and not to waste your time.

C. My friend advised me not to sit idle and not to waste my time.

D. My friend eulogised me not to sit idle and not to waste my time.

E. My friend ordered me not to sit idle and not to waste your time.

42. Choose the correct change of narration from the options:
Nakul said to me, "Your project may be very effective but you will not be able to convince Mr. Nikhil Agrawal."
A. Nakul told me that your project may be very effective but you will not be able to convince Mr. Nikhil Agrawal.
B. Nakul told me that my project might be very effective but I will not be able to convince Mr. Nikhil Agrawal.
C. Nakul told me that your project may be very effective but you would not be able to convince Mr. Nikhil Agrawal.
D. Nakul told me that my project may be very effective but I would not be able to convince Mr. Nikhil Agrawal.
E. Nakul told me that my project might be very effective but I would not be able to convince Mr. Nikhil Agrawal.

43. Choose the correct change of narration from the options:
Our teacher said to me, "This is your mistake."
A. Our teacher told me that this was your mistake.
B. Our teacher told me that that was my mistake.
C. Our teacher told me that was your mistake.
D. Our teacher told me that that is my mistake.
E. Our teacher told me that this was my mistake.

44. Find out the idiom for the following:
A distorted information or rumour/ gossip about something.
A. Chinese proverb
B. rumour
C. gossip
D. Chinese whispers
E. falsehood

45. Choose the correct option.
He _____me _____my project in due time.
A. has help, wrote
B. help, write

C. helps, written
D. helping, to wrote
E. helped, write

46. Choose the correct option.
Our professor _____to Cambridge last year.
A. visited at
B. went
C. had been gone
D. gone
E. has been gone

47. Choose the correct option:
What will you do if you _____ the history exam?
A. would fail
B. will fail
C. fail
D. failed
E. fails

48. Locate which part is erroneous, if any:
Subjected to humiliation, (A)/ the producers of the reality television show (B)/ should have anticipated (C)/ the contestant might break the contract. (D)/ No Error (E)

49. Which sentence has an example of misplaced modifier?
A. At the lake, Ajay caught a fish that was very small.
B. To win the race, much determination is needed.
C. Yesterday she promised to call me.
D. Karl Benz has, by most automobile historians, been given credit for the invention of the automobile.
E. None of the above sentences has a misplaced modifier.

50. Choose the most logical order of following sentences:

A. Luckily the tide of battle moved elsewhere after the American victory at Midway and an Australian victory over Japan at Milne Bay.

B. It could have been no more than a delaying tactic.

C. The Australian military, knowing the position was hopeless, planned to fall back to the south-east in the hope of defending the main cities.

D. They had captured most of the Soloman Islands and much of New Guinea, and seemed poised for an invasion.

E. Not many people outside Australia realize how close the Japanese got.

A. EDCBA

B. ECDAB

C. ADCBE

D. CDBAE

E. ABCDE

51. Choose the correct change of narration from the options:

He said to me, "Why have you come here?"

A. He asked me why I had come here.

B. He asked me why I had gone there.

C. He asked me why I had come there.

D. He asked me if I had gone here.

E. He asked me whether I had come here.

52. Choose the correct change of narration from the options:

Priyanka says, "I do not work as a freelancer."

A. Priyanka says that she does not work as a freelancer.

B. Priyanka says that she did not work as a freelancer.

C. Priyanka said that she did not worked as a freelancer.

D. Priyanka said that she does not work as a freelancer.

E. Priyanka says that she never worked as a freelancer.

53. Find out the idiom for the following:

Learn to do something with expertise.

A. learn the details

B. knows the job

C. learn the ropes
D. expert in field
E. mastery

54. Choose the correct option.
Ram _____his appointment _____next week.
A. got, schedule
B. get, has scheduled
C. get, is scheduling
D. got, scheduled
E. gotten, schedule

55. Select the appropriate brief expression for the following:
Agreeing on something which is being done or suggested.
A. on the same page
B. on the same line
C. agree
D. sharing same thought
E. optimistic

56. Choose the correct alternative for the underlined parts of the sentence.
My teacher made me read five novels in twenty days.
A. make, reads
B. has make, read
C. making, is reading
D. make, reading
E. no correction required

57. Find out the idiom for the following:
Someone who is good at calculating numbers.
A. number expert
B. number cruncher
C. accountant
D. mathematician
E. numismatic

58. Choose the correct option.
I _____the electrician_____at my broken light.
A. having, look
B. had, been looked
C. had, look
D. have, is looking
E. has, look

59. Choose the correct option.
As soon as the old clock on the town hall _____ midnight, it _____quiet in the street.
A. struck, becames
B. is struck, became
C. was struck, becoming
D. has strucks, become
E. struck, became

60. Choose the correct option:
You will catch the train if you _____ earlier.
A. left
B. leaves
C. leave
D. would leave
E. had left

61. Locate which part is erroneous, if any:
The pedestrian (A)/ was hit (B)/ by a car (C)/, sitting on the curb. (D)/ No Error (E)

62. Locate which part is erroneous, if any:
He struck (A)/ the fishbowl (B)/ with his forehead, (C)/ which fortunately was empty. (D)/ No error.

63. Choose the most logical order of following sentences:

A. To much of the Labour movement, it symbolizes the brutality of the upper classes.
B. And to everybody watching, the current mess over foxhunting symbolizes the government's weakness.
C. To foxhunting's supporters, Labour's 1991 manifesto commitment to ban it symbolizes the party's metropolitan roots and hostility to the countryside.
D. Small issues sometimes have large symbolic power.
E. To those who enjoy thundering across the countryside in red coats after foxes, foxhunting symbolises the ancient roots of rural lives.

A. DEACB
B. ECDBA
C. CEADB
D. DBAEC
E. BEADC

64. Choose the correct change of narration from the options:
I said to Rajat, "Let us play football."
A. I suggested Rajat that let us play football.
B. I suggested Rajat that I should play football.
C. I suggested Rajat that let me play football.
D. I suggested Rajat that he should play football.
E. I suggested Rajat that we should play football.

65. Choose the correct change of narration from the options:
She says, "I study everyday in the morning."
A. She says that she studies everyday in the morning.
B. She says that she studied everyday in the morning.
C. She says that she study everyday in the morning.
D. She will say that she studies everyday in the morning
E. She says that I study everyday in the morning.

66. Find out the idiom for the following:
An employee is paid the necessary minimum salary.
A. paid peanuts
B. less wages employee
C. underpaid employee
D. paid nuts
E. bonded labour

67. Choose the correct option.
My mother never _____us _____ food with dirty hands.
A. let, eaten
B. let, eat
C. letting, eats
D. let, to ate
E. let, eating

68. Choose the correct option.
If you_____ him, he _____helped you.
A. request, will be
B. requesting , would
C. have request, would have
D. requested, will has
E. had requested, would have

69. Choose the correct option:
If I eat peanut butter, I _____ sick.
A. would have gotten
B. would get
C. get
D. got
E. gets

70. Locate which part is erroneous, if any:

The no-duty rule(A)/ should not apply (B)/ to a patron struck by a ball (C)/ batted by a player during the seventh inning stretch (D)/ while seated in an open concourse. (E)

71. Choose the sentence that does NOT include a dangling modifier.
A. Being a sit-down comedian, it was hard to make ends meet.
B. Biking to class, the sun shined upon the student.
C. Sitting on the threshold, a snake bit him.
D. Devouring ample amounts of cookout, the engineer felt joy seeping back into his soul.
E. All the above sentences have dangling modifiers.

72. Choose the most logical order of following sentences:
A. This feeling of an extensive group gives rise to a fellow feeling, a feeling of brotherhood amongst the citizens.
B. This feeling takes up beyond the bounds of family, caste, religion as well as region and helps us develop a broad perspective that all of us together constitute an extensive group called 'the nation'.
C. National integration is the feeling among all the citizens of a country that they all are part of one nation.
D. We do not, then, limit our thinking to our own caste or religion but think about all our fellow citizens.
A. CDAB
B. CABD
C. CDBA
D. CBDA
E. CBAD

73. Choose the correct change of narration from the options:
She said to me, "Let us visit the market".
A. She suggested me that they should visit the market.
B. She suggested me that I should visit the market.
C. She suggested me that we should visit the market.
D. She suggested me that we would visit the market.
E. She suggested me that we shall visit the market.

74. Choose the correct change of narration from the options:
I said to my mother, "I will take only a cup of tea because I am not hungry".
A. I told my mother that I will take a cup of tea because I was not hungry.
B. I told my mother that I would take a cup of tea because I am not hungry.
C. I told my mother that I would take a cup of tea because I was not hungry.
D. I told my mother that I will take a cup of tea because I am not hungry.
E. I told my mother that I would have taken a cup of tea because I was not hungry.

75. Choose the correct alternative for the underlined parts of the sentence.
I will have my secretary to sent you an email regarding the placement details.
A. will have, send
B. would have, sends
C. will have, is sending
D. would have, sending
E. no correction required

76. Choose the correct change of narration from the options:
Payal said to Monika, "She is better than she was".
A. Payal told Monika that she is better than she had been.
B. Payal told Monika that she was better than she was.
C. Payal told Monika that she was better than she has been.
D. Payal told Monika that she was better than she had been.
E. Payal told Monika that she is better than she was.

77. Choose the correct change of narration from the options:
I said to my servant, "At what time did you go to the station yesterday?"
A. I asked my servant at what time he went to the station the previous day.
B. I asked my servant at what time he had gone to the station yesterday.
C. I asked my servant at what time he had gone to the station the previous day.
D. I asked my servant at what time he had gone to the station that day.
E. I asked my servant at what time he had gone to the station the next day.

78. Choose the most logical order of following sentences:

A. Since then, intelligence tests have been mostly used to separate dull children in school from average or bright children, so that special education can be provided to the dull.

B. In other words, intelligence tests give us a norm for each age.

C. Intelligence is expressed as intelligence quotient, and tests are developed to indicate what an average child of a certain age can do: what a 5-year-old can answer, but a 4-year-old cannot, for instance.

D. Binet developed the first set of such tests in the early 1900s to find out which children in school needed special attention.

E. Intelligence can be measured by tests.

A. CDABE

B. DECAB

C. ECDAB

D. CBADE

E. ABCDE

79. Select the correct response for the sentence: "Being open to the sun, the windowsill is the proper place for an indoor man's plants."

A. This sentence contains a dangling modifier. The correct subject need to be inserted after this dangling modifier to correct it.

B. There is no error and the sentence should remain unchanged.

C. The sentence contains a misplaced modifier. Consider reorganizing the phrase: "indoor man's plants".

D. This sentence contains squinting modifier.

E. None of the responses is correct.

80. Choose the correct alternative for the underlined parts of the sentence.

Did your boss <u>been have</u> you <u>worked</u> late last night?

A. had, works

B. has, worked

C. have, is working

D. have, work

E. no correction required

81. Choose the correct change of narration from the options:
He said to me, "Honesty is the best policy."
A. He told me that honesty was the best policy.
B. He told me that honesty should be the best policy.
C. He tells me that honesty is the best policy.
D. He told me that honesty is the best policy.
E. He says that honesty is the best policy.

82. Select the appropriate brief expression for the following:
Tasks which are easily achieved.
A. easy job
B. low-hanging fruit
C. mean job
D. handsome job
E. probability of happening

83. Choose the correct alternative for the underlined parts of the sentence.
His reaction did forced us to rethinking our policy.
A. forced, to rethink
B. forcing, to rethinked
C. force, to rethought
D. had been forced, think
E. no correction required

84. Choose the correct alternative for the underlined part of the sentence.
When he came in, I write a letter.
A. writing
B. was writing
C. is writing
D. writes
E. no correction required

85. Choose the appropriate answer:

Rafael Nadal _____ his tie with his greatest rivals, Roger Federer and Novak Djokovic.

A. break
B. broke
C. broken
D. have broken
E. has been breaking

86. Choose the correct option:
I have no fire insurance. If my house _____ down, I would have no way to buy new furniture.
A. burns
B. burned
C. had burned
D. has burned
E. None of these

87. Locate which part is erroneous, if any:
The salesman (A)/ tried (B)/ to sell (C)/ a car to a customer (D)/ that was broken down (E).

88. In the sentence that follows, a section has been underlined. You are required to replace it with the correct idiom.
Examining job applications is the first stage of examining a group of people and decide who are suitable and who are not.
A. examining the cats and dogs
B. examining the real problem
C. examining the bulls and bears
D. separating the cats and dogs
E. separating the sheep from the goats

89. Choose the most logical order of following sentences:

A. It also gives rise to a feeling of animosity among the different sections of the society.

B. In a democratic system, frequent use of power is never desirable, be it on the part of government or the people.

C. Therefore, citizens should never resort to violent ways and means in democracy, though they have the right to oppose the government.

D. It destroys the stability and security in public life.

A. DBAC

B. BDCA

C. BDAC

D. DACB

E. DCBA

90. Choose the correct change of narration from the options:

Abhishek said to him, "Your team will win you the first prize."

A. Abhishek told him that his team would win him the first prize.

B. Abhishek told him that his team will win him the first prize.

C. Abhishek told him that your team would win you the first prize.

D. Abhishek told him that my team would win me the first prize.

E. Abhishek told him that your team will win you the first prize.

Course Name: Database Management System

Course Outcome

CO1: Understand the concept of database management systems and Relational database.
CO2: Identify the various data model used in database design.
CO3: Design conceptual models of a database using ER modeling for real life applications and construct queries in Relational Algebra.
CO4: Create and populate a RDBMS for a real-life application, with constraints and keys using RA.
CO5: Select the information from a database by formulating complex queries in RA.
CO6: Analyze the existing design of a database schema and apply concepts of normalization to design an optimal database.
CO7: Discuss indexing mechanisms for efficient retrieval of information from a database.
CO8: Discuss recovery system and be familiar with introduction to web database, distributed databases.

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End Term Examination, Even Semester 2021-22

B.Tech (CSE), II, IV Sem-(CSE ALL)

Subject Code & Subject Name- BCSC0003 Database Management System

Time: 3 Hours **Maximum Marks: 50**

Instruction for students:

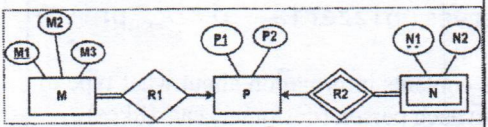
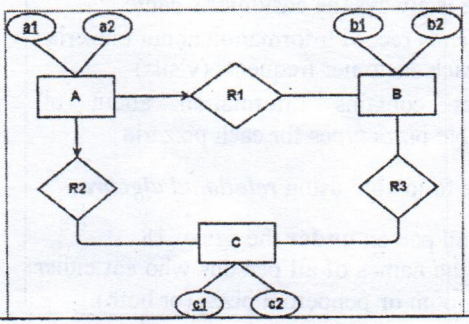
1. Attempt all questions
2. don't switch between the section A, B and Section C, first solve one section then move to second
3. Use only blue pen to answer
4. No B sheet will be given

Section – A

*Attempt **All** Questions* 4 X 5 = 20 Marks

No.	Detail of Question	Marks	CO	BL	KL

1	Consider the following schema:
	`Person(name, age, gender)`
	`Frequents(name, pizzeria)`
	`Eats(name, pizza)`
	`Serves(pizzeria, pizza, price)`
	Description:
	• Eats contains information about what type of pizza each person (customer) eats (likes)
	• Name is always the customer's name
	• Frequents record information about pizzerias that each customer frequents (visits)
	• Serves contains information about all possible pizza types for each pizzeria
	
	Answer the following using ***relational algebra***:
	
	a. Find all person **under** the age of 18.
	b. Find the names of all persons who eat either mushroom **or** pepperoni pizza (or both).
		4	CO5	C	P

	c. Find the names of all persons who eat both mushroom **and** pepperoni pizza. d. Find all pizzerias that where Amy eats.				
2	Please create **an Entity Relationship diagram** that captures the information about the College DB. Be certain to indicate identifiers and cardinality constraints. • A college contains many departments • Each department can offer any number of courses • Many instructors can work in a department • An instructor can work only in one department • For each department there is a Head • An instructor can be head of only one department • Each instructor can take any number of courses • A course can be taken by only one instructor • A student can enroll for any number of courses • Each course can have any number of students	4	CO3	C	P
3	Convert into minimum number of relations required for the following ER diagram in the relational model a.  b.  	4	CO2	AN	P

4	Answer the following: a. Which normal form is considered adequate for normal relational database design? b. Consider a schema R(A, B, C, D) and functional dependencies A→B and C→D. Test and tell whether the decomposition of R into R1 (A, B) and R2(C, D) is dependency preserving or not? c. Is this decomposition in part (b) is lossy? (with reason)	4	CO6	AN	C
5	Write short note on any *three* of the following: a. MVD and 4NF b. JD and 5NF c. Trivial and non-Trivial FDs d. Draw three Schema architecture. Explain it briefly.	4	CO6	R	F

Section – B

*Attempt **All** Questions* 3 X 5 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
1	Discuss Anomalies with help of examples.	3	CO7	F	C
2	Find the minimal cover of the following FDs: a. $F: \{A \rightarrow C, AB \rightarrow C, C \rightarrow DI, CD \rightarrow I, EC \rightarrow AB, EI \rightarrow C\}$ b. $F: \{PQ \rightarrow R, PR \rightarrow Q, Q \rightarrow S, QR \rightarrow P, PQ \rightarrow T\}$ c. $F: \{V \rightarrow W, VW \rightarrow X, Y \rightarrow VX, Y \rightarrow Z\}$	3	CO6	U	C
3	Define the following with an example: a. BCNF b. 3NF c. 2NF	3	CO7	U	C
4	Suppose we have a relation $R(X,Y,Z,W)$ with the FD's: $X \rightarrow Y, Y \rightarrow Z, Z \rightarrow W$ Which one of the following decompositions is not lossless? a. $R_1(X,Y), R_2(Y,Z), R_3(Z,W)$ b. $R_1(X,Y), R_2(X,Z), R_3(X,W)$ c. $R_1(X,W), R_2(Y,W), R_3(Z,W)$	3	CO6	AN	P

5	Answer the following: a. What is the difference between wait-die and wound-wait algorithms? b. Suppose that transaction T_5, T_{10}, T_{15} have time-stamps 5, 10 and 15 respectively. i. What will happen to T_5 requesting for data item holds by T_{10} in *wait-die* scheme? ii. What will happen to T_{15} requesting for data item holds by T_{10} in *wait-die* scheme? iii. What will happen to T_5 requesting for data item holds by T_{10} in *wound-wait* scheme? iv. What will happen to T_{15} requesting for data item holds by T_{10} in *wound-wait* scheme?	3	CO5	A	P

Section – C

*Attempt **All** Questions* 5 X 3 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
1	Answer the following: a. Discuss Properties of the transaction b. State diagram of the transaction	5	CO7	U	C
2	Answer the following: a. Discuss two-phase locking protocol. What is the limitation of 2PL? How strict 2PL helpful in overcoming those limitations? What is a locking point? b. What is Distributed database? Discuss fragments and its type with help of example?	5	CO8	U	P
3	Answer the following: **a.** Which of the following schedules is conflict serializable? Explain why or why not the schedule is serializable. For each serializable schedule, determine the equivalent serial schedule. i. $r_1(X); r_3(X); w_1(X); r_2(X); w_3(X);$ ii. $r_1(X); r_3(X); w_3(X); w_1(X); r_2(X);$ iii. $r_3(X); r_2(X); w_3(X); r_1(X); w_1(X);$	5	CO7	E	M

	iv. $r_3(X); r_2(X); r_1(X); w_3(X); w_1(X);$ **b.** Consider schedules S3, S4, and S5 below. Determine whether each schedule is strict, cascadeless, recoverable, or nonrecoverable. (Determine the strictest recoverability condition that each schedule satisfies.) **S3**: $r_1(X); r_2(Z); r_1(Z); r_3(X); r_3(Y); w_1(X); c_1;$ $w_3(Y); c_3; r_2(Y); w_2(Z); w_2(Y); c_2;$ **S4**: $r_1(X); r_2(Z); r_1(Z); r_3(X); r_3(Y); w_1(X); w_3(Y);$ $r_2(Y); w_2(Z); w_2(Y); c_1; c_2; c_3;$ **S5**: $r_1(X); r_2(Z); r_3(X); r_1(Z); r_2(Y); r_3(Y); w_1(X);$ $c_1; w_2(Z); w_3(Y); w_2(Y); c_3; c_2;$				

Course Name: Operating Systems

Course Outcome

CO1- Understand the classification of operating system environment.
CO2- Understand the basic of process management.
CO3- Apply the concept of CPU process scheduling for the given scenarios.
CO4- Illustrate the process synchronization and concurrency process in operating system.
CO5- Analyze the occurrence of deadlock in operating system.
CO6- Describe and analyze the memory management and its allocation policies.
CO7- Understand the concepts of disk scheduling

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End Term Examination, Even Semester 2021-22
B. Tech. (CSE), II Year, IV Semester
Operating Systems (BCSC0004)

Time: 3 Hours **Maximum Marks: 50**

Section – A

Attempt All Questions 4 X 5 = 20 Marks

No.	Detail of Question	Marks	CO	BL	KL
1	State the Dining Philosopher problem. Discuss the solution to Dining Philosopher problem using semaphore.	4	4	U	C
2	Consider the following table of arrival time and burst time for three processes P_0, P_1 and P_2. Process | Arrival time | Burst time P_0 | 0 ms | 9 ms P_1 | 1 ms | 4 ms P_2 | 2 ms | 9 ms Scheduling is carried out only at arrival or completion of a process. Calculate the average waiting time for pre-emptive shortest job first scheduling algorithm for the three processes?	4	3	A	P
3	Discuss all the services provided by the Operating System to both the users and to the programs.	4	1	U	C
4	What are the necessary conditions for a deadlock to arise? Explain them.	4	5	R	F

5	Suppose the following disk request sequence (track numbers) for a disk with 100 tracks is given: 45, 20, 90, 10, 50, 60, 80, 25, 70. Assume that the initial position of the R/W head is on track 50. Calculate the additional distance that will be traversed by the R/W head when the Shortest Seek Time First (SSTF) algorithm is used compared to the SCAN (Elevator) algorithm (assuming that SCAN algorithm moves towards 100 when it starts execution).	4	7	A	P

Section – B

Attempt All Questions 3 X 5 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
1	What is safe state in deadlock? A system shares 9 tape drives. The current allocation and maximum requirement of tape drives for three processes are shown below: Process | Current Allocation | Maximum Requirement P1 | 3 | 7 P2 | 1 | 6 P3 | 3 | 5 Determine whether the current state of the system is safe or not?	3	5	A	P
2	Discuss the advantages of Dynamic Partitioning over fixed partitioning memory management technique.	3	6	U	C
3	Consider a computer system with 40-bit virtual addressing and page size of sixteen kilobytes. If the computer system has a one-level page table per process and each page table entry requires 48 bits, then calculate the size of the per-process page table in megabytes.	3	6	An	M

4	What are the various attributes that are associated with an opened file? Explain the single level directory structure with its advantages and disadvantages.	3	7	R	F
5	Distinguish between Logical and Physical address. A paging scheme uses a Translation Lookaside buffer (TLB). The effective memory access takes 190 ns and a main memory access takes 50 ns. What is the TLB access time (in ns) if the TLB hit ratio is 90% and there is no page fault?	3	6	An	C

Section – C

Attempt All Questions 5 X 3 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
1	Consider the following snapshot of the system: Max | Allocation | Available A B C | A B C | A B C P0 0 0 1 | 0 0 1 | 1 5 2 P1 1 7 5 | 1 0 0 | P2 2 3 5 | 1 3 5 | P3 0 6 5 | 0 6 3 | Find out the following by applying Banker's Algorithm: a) How many resources are there of type (A, B, C)? b) What are the contents of the Need matrix? c) Is the system in a safe state? Why?	5	5	A	P
2	What is Thrashing? Discuss the effects and techniques to handle the Thrashing.	5	6	R	C
3	Explain the Demand Paging. Consider a main memory with five-page frames and the following sequence of page references: 3, 8, 2, 3, 9, 1, 6, 3, 8, 9, 3, 6, 2, 1, 3. Calculate the page fault for the First-In-First Out (FIFO), Least Recently Used (LRU) and Optimal page replacement policies?	5	6	An	M

Course Name: Cloud and Business Process Management.

Course Outcome

CO1- Understand basics of cloud Storage systems.
CO2- Explain the technologies and approaches for the business related issues.
CO3- Understand the operation view and service catalog of cloud management.
CO4- Understand the concepts of VPM cloud computing.
CO5- Design process interactions interface for business users.

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End Term Examination, Even Semester 2021-22
B- Tech (CCV), Year- II, Semester- IV
BCSE 0508 Cloud and Business Process Management

Time: 3 Hours **Maximum Marks: 50**

Instruction for students:
1) Strictly follow the question no. sequence while writing answers.

Section – A

*Attempt **All** Questions* 4 X 5 = 20 Marks

No.	Detail of Question	Marks	CO	BL	KL
1	Explain Smart metering architecture in detail along with the proper description of various metering and billing policies.	4	1	U	F
2	An organization wants to go with BPM. Outline the benefits that BPM provides to the organization. Also, discover how BPM helps in improving business strategy?	4	2	An	C
3	Explain BPM life cycle with the appropriate description of the evaluation of the BPM lifecycle phase.	4	4	U	F
4	Identify the challenges of business process reengineering. Discuss the important factors in BPR along with various phases involved in this process.	4	3	Ap	P
5	What is IBM Cloud Computing Reference Architecture? Explain with the help of Diagrammatic representation. Also explain shared system capabilities.	4	5	R	F

Section – B

*Attempt **All** Questions* 3 X 5 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
1	With reference to business process automation, design the procedure that needs to be followed to upload 10 files inside a directory to docker hub by applying docker commit operation. Write the appropriate steps with description.	3	5	C	D
2	An organization has decided to make its cloud solution resilient. Recommend how this organization can achieve resiliency its cloud solution? Also, explain about the support system capabilities?	3	3	E, R	M
3	Differentiate between coach and Coach View in IBM BPM? Write short note on heritage coaches.	3	1	An, R	C
4	Explain in detail Polling agent, Resource agent, and Monitoring agent.	3	1	U	F
5	What is "BPM-enabled process automation"? Write short notes on the Degree of repetition and degree of automation.	3	2	R	F

Section – C

*Attempt **All** Questions* 5 X 3 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
1	In reference of level of abstraction, classify and explain types of abstraction along with the diagrammatic representation.	5	5	An, R	C
2	Identify the type of services that can be hosted inside the Aneka container. List the features of Aneka that provides scalability of the resource.	5	4	Ap, R	C
3	Distinguish between different models of self-service provisioning. Explain in detail the Inter- cloud Operation framework.	5	1	An, R	M

CO – Course Outcome, BL – Abbreviation for Bloom's Taxonomy Level (R-Remember, U-Understand, A-Apply, An-Analyze, E-Evaluate, C-Create), KL – Abbreviation for Knowledge Level (F-Factual, C-Conceptual, P-Procedural, M-Metacognitive). However, For Engg. Courses in addition to F, C, P & M include D-Fundamental Design Principles, S-Criteria and Specifications, PC-Practical Constraints, DI- Design Instrumentalities

Course Name: Applied Statistical Analysis

Course Outcome

CO1-Understand the different variables measurement and scalling.
CO2-Analyze the different central tendency operations.
CO3-Demonstrate the different ways to represent the data.
CO4-Analyze the hypothesis testing using different estimation and distribution methods.
CO5-Analyze the correlation and regression for the real world problem data
CO6-Analyze the data using the advanced techniques like, Chi-squared goodness-of-fit test, Chi-Square test of independence, the sign test and Mann-Whitney test.

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End Term Examination, Even Semester 2021-22
B.Tech (CSE(DA)), IInd Year, IVthSemester
BCSE0552 &Applied Statistical Analysis

Time: 3 Hours **Maximum Marks: 50**

Instruction for students:

1.Write your Roll No. on the top immediately on receipt of this question paper
2. Attempt all the questions.

Section – A

Attempt All Questions 4 X 5 = 20 Marks

No.	Detail of Question	Marks	CO	BL	KL
1	An investor is fond of investing in equity shares. During a period of falling prices in the stock exchange, a stock is sold at Rs 120 per share on one day, Rs 105 on the next and Rs 90 on the third day. The investor has purchased 50 shares on the first day, 80 shares on the second day and 100 shares on the third day. What average price per share did the investor pay?	4	2	E	C
2	You have been given some raw data and asked to present them in a suitable frequency distribution with some class-intervals. Explain what factors you would consider in deciding the number of classes and the lower and upper limits of class-intervals for the proposed frequency distribution.	4	3	A	C
3	A perfect die is thrown a large number of times in sets of 8. The occurrence of 5 or 6 is called a success. In what proportion of the sets would you expect three successes?	4	4	E	C
4	An auto company claims that the petrol consumption of the new car introduced by it is 9.5 km per litre. In an experiment with 50 cars, it was found that the	4	4	E	C

	mean petrol consumption was 10 km per litre, with a standard deviation of 3.5 km per litre. Test the validity of the claim of the car company using 5% level of significance (5% level of significance for one-tail test is 1.64.).				
5	The demand for a particular spare part in a factory was found to vary from day-to-day. In a sample study, the following information was obtained. Test the hypothesis that the number of parts demanded does not depend on the day of the week. Test this at 5% level.(Required Data =11.070)	4	6	A	P

Day	Mon	Tue	Wed	Thu	Fri	Sat
Number of parts demanded	1124	1125	1110	1120	1126	1115

Section – B

Attempt All Questions 3 X 5 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
1	Define correlation? Do you think that correlation always signifies a cause and effect relationship between the two variables?	3	5	U	F
2	Describe the chi-square distribution. What are the parameters on which the shape of a chi-square distribution curve depends?	3	6	U	C
3	A company is engaged in the manufacture of car tyres. Their mean life is 42,000 km with a standard deviation of 3,000 km. A change in the production process is believed to improve the quality of tyres. A test sample of 28 new tyres has a mean life of 43,500 km. Do you think that the new car tyres are significantly superior to the earlier one? Test the hypothesis at 5 per cent level of significance.(Required data: 1.703)	3	4	An	P
4	In 1983, 15 per cent of households in a certain city indicated that they owned a sewing machine. In 1985, there was a reason to believe that there was some increase in its percentage. A survey based on random sample of 900 households was taken and it	3	4	An	P

	was found that 189 households had a sewing machine. Can we conclude that there has been a significant increase in the sale of sewing machines, assuming 0.05 level of significance?(Required data: 1.64)				
5	Define F-test? Discuss its application in testing whether the two variances are homogeneous.	3	4	U	C

Section – C

*Attempt **All** Questions* 5 X 3 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
1	A company has devised three training methods to train its workers. It is keen to know which of these three training methods would lead to greatest productivity after training. Given below are productivity measures for individual workers trained by each method. Find out whether the three training methods lead to different levels of productivity at the 0.05 level of significance. (Required data: 3.55) Method 1: 30, 40, 45, 38, 48, 55, 52 Method 2: 55, 46, 37, 43, 52, 42, 40 Method 3: 42, 38, 49, 40, 55, 36, 41	5	6	An	P
2	Find the coefficient of correlation between X and Y, given that, $10Y = 8X + 66$ and $40X = 18Y + 214$ are the regression of Y on X and X on Y, respectively.	5	5	E	C
3	A potential buyer wants to decide which of the two brands of electric bulbs he should buy as he has to buy them in bulk. As a specimen, he buys 100 bulbs of each of the two brands A and B. On using these bulbs, he finds that brand A has a mean life of 1,200 hours with a standard deviation of 50 hours and brand B has a mean life of 1,150 hours with a standard deviation of 40 hours. Do the two brands differ significantly in quality? Use a = 0.05. (Required data: 1.96)	5	4	E	C

Course Name: IT NETWORK SECURITY

Course Outcome

CO1: Understand the fundamentals of network security.

CO2: Demonstrate the principle of network security controls, protocols and devices

CO3: Understand the basic concepts of physical and host security, and secure IDS configuration & Management.

CO4: Explain the network security by using VPN and wireless technologies.

CO5: Explain the concepts of network traffic monitoring and analysis.

CO6: Conceptualize the risk management framework and vulnerability management.

CO7: Explain the concepts of data backup, recovery tools and incident response management.

Printed Pages: 3 **University Roll No. .....**

End Term Examination, Even Semester 2021-22

Program: B.Tech (CSF) Year: II Semester: IV

Subject Code & Subject Name- IT NETWORK SECURITYBCSE 0602

Time: 3 Hours **Maximum Marks: 50**

Section – A

Attempt AllQuestions 4 X 5 = 20 Marks

No.	Detail of Question	Marks	CO	BL	KL
1	Explain with duties various servers of Kerberos. Illustrate the sequence of message exchanges that happens when a client attempts to obtain a service granting ticket in reference to Kerberos 4 by detailing each step.	4	2	An	C
2	What do you understand by Server security? Discuss and explain three major components which require hardening of web server with suitable diagram.	4	1	R	F
3	Why is SSL (Secure Socket Layer Protocol) layer positioned between the application layer and transport layer? Discuss the following sub-protocol of SSL. I. Handshake protocol II. Record protocol III. Alert protocol	4	2	U	C
4	Write down various Quality of Service (QoS) parameters in VPNs. Discuss various types of VPN	4	4	U	U

	briefly.				
5	As A Risk Manager, How Can You Ensure Monitoring and Control of Risks? Also Briefly Explain the Process of Risk Management. Or What is Incident handling & response (IH & R) processes? Discuss and explain Incident Handling and Response Process with suitable example.	4	6	U	F

Section – B

Attempt All Questions 3 X 5 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
1	What are the various Factors that could influence Internet speed while using a VPN service and the technique to improve the speed of a VPN.	3	4	U	D
2	Wireless networks uses IEEE standard of 802.11 and uses radio waves for communication what are the Advantages and Limitations of wireless network over the wired network.	3	4	R	P
3	DiscussNetwork Traffic signatures? A signature is a set of characters that define network activity so how these signatures are useful to detecting Suspicious activity and Attack Signatures Analysis.	3	5	An	M
4	"The risk management frameworks require organizations to maintain a vulnerability management program". Justify your answer with suitable example.	3	6	A	D
5	If computer A needs a data from computer B, it will need a copy of data from the server, to which computer B is connected. This can be done through, file transfer, inter-process communication and backup. Even though data is transferred from computer B to computer A, there can be a probability that computer A may face the situation of untimely data errors, an expensive transferring process taking place b/w two servers or any other operational process. How you will resolve this issue with SAN architecture.	3	7	E	M

Section – C

*Attempt **All** Questions* 5 X 3 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
1	Differentiate between Trusted VPNs and Secure VPNs. Discuss and explain Point-to-Point VPN Topology and Full Mesh VPN Topology with suitable example and diagram.	5	4	An	P
2	Discuss the various steps of Deploying a Wireless IDS (WIDS) and a Wireless IPS (WIPS) with suitable diagram.	5	3,4	A	C
3	Explain the Redundant Array of Independent Disks(RAID) Storage Architecture with its major components. Also Advantages/ disadvantages of RAID systems.	5	7	U	F

Course Name: Algorithms for Intelligent Systems and Robotics

Course Outcome

CO1- Understand the Theoretical aspect of various searching strategies to solve AI Problems.
CO2- Under a range of techniques of intelligent systems across artificial intelligence and intelligent Agents.
CO3- Describe and Apply various techniques for logic programming and Machine Learning.
CO4-Understand the Practical Implementation of various tools for the implementation of Robotics.

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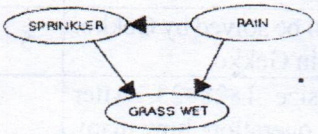
End Term Examination, Even Semester 2021-22
B.Tech (CSE-AIML), II Year, IV Semester
BCSE 0702 : Algorithms for Intelligent Systems and Robotics

Time: 3 Hours **Maximum Marks: 50**

Instruction for students:
1. Don't switch between the section A, B and Section C, first solve one section then move to second.
2. Use only blue pen to answer
3. No B sheet will be given

Section – A

*Attempt **All** Questions* 4 X 5 = 20 Marks

No.	Detail of Question	Marks	CO	BL	KL
1	What are well-defined problems? Illustrate the vacuum world state space graph. Identify states, actions, goal test, and path cost.	4	CO1	U	F
2	What is the probability that it is raining, given the grass is wet? SPRINKLER RAIN | T | F F | 0.4 | 0.6 T | 0.01 | 0.99  RAIN T | F 0.2 | 0.8 GRASS WET SPRINKLER RAIN | T | F F F | 0.0 | 1.0 F T | 0.8 | 0.2 T F | 0.9 | 0.1 T T | 0.99 | 0.01	4	CO2	C	P
3	Draw the inference of the following using forward chaining. (i) The law says that it is a crime for a Gaul to sell potion formulas to hostile nations. (ii) The country Rome, an enemy of Gaul, has acquired some potion formulas, and all of its formulas were sold to it by Druid. (iii) Druid is a Gaul. Is Druid a criminal?	4	CO2	U	P

4	How ROS Environment is set up? Explain CATKIN workspace.	4	CO4	A	C
5	Explain Backpropagation algorithm for learning of an ANN.	4	CO4	U	C

Section – B

*Attempt **All** Questions* 3 X 5 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
1	Solve the following differential equation with initial condition y(0) = 5 using Gekko **k dy/dt = - t y** where k=10 The solution of y(t) should be reported from 0 to final time 10.	3	CO3	F	C
2	Define different layers of the Convolutional Neural Network.	3	CO3	U	C
3	Differentiate ANN and CNN with Diagram.	3	CO3	U	C
4	What are the services offered by ROS and OS.	3	CO4	U	P
5	What is the need of ROS? Explain the ROS File System.	3	CO4	U	P

Section – C

*Attempt **All** Questions* 5 X 3 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
1	**a.** Discuss how linear equations will be solved by Gekko. **b.** Enlist different solver available in Gekko.	5	CO3	U	C
2	**a.** Consider an input image of size 1×28×28. After applying following convolution operation, how many parameters for each filter will be there: model2.add (convolution2D, filter (3,3), input_shape(28×28×1)) Also find the size of an output image if stride size is 2. **b.** What is the size of an output image after applying the following syntax? model2.add (Maxpooling2D(2,2))	5	CO3	U	P
3	Write short notes on: a. ROS Core c. ROS Topics b. ROS Node d. ROS Messages	5	CO4	U	C

Printed Pages: 15 **University Roll No.....**

End-Term Examination, Even Semester 2021-22

B.Tech. CSE (IIOT) & CSE, II & IV: Year, IV & VIII: Semester

BCSE 0903: Data Science and Analytics

Time: 1.5-hour **Maximum Marks: 50**

Section A

Note: Attempt all questions. **30 x 1 = 30 Marks**

1. Which of the following libraries allows to manipulate, transform and visualize data easily and efficiently.
   A. Pandas
   B. NumPy
   C. Matplotlib
   D. All of the above

2. PANDAS stands for _____
   A. Panel Data Analysis
   B. Panel Data analyst
   C. Panel Data
   D. Panel Dashboard

3. _____ is an important library used for analyzing data.
   A. Math
   B. Random
   C. Pandas
   D. None of the above

4. Important data structure of pandas is/are _____
   A. Series
   B. DataFrame
   C. Both of the above
   D. None of the above

5. Which of the following command is used to install pandas?

A. pip install pandas
B. install pandas
C. pip pandas
D. None of the above

6. Which of the following command is used to import pandas?
A. import pandas as pd
B. import pandas
C. from pandas import *
D. All of above

7. A _____ is a one-dimensional array.
A. Data Frame
B. Series
C. Both of the above
D. None of the above

8. Which of the following statement is wrong?
A. We can create Series from Dictionary in Python.
B. Keys of dictionary become index of the series.
C. Order of indexes created from Keys may not be in the same order as typed in dictionary.
D. All are correct

9. What will be output for the following code?

```
import numpy as np

a = np.array([1,2,3])

print a
```

A. [[1, 2, 3]]
B. [1]
C. [1, 2, 3]
D. Error

10. What will be output for the following code?

```
import numpy as np
a = np.array([1, 2, 3], dtype = complex)
print a
```

A. [[1.+0.j, 2.+0.j, 3.+0.j]]
B. [1.+0.j]
C. Error
D. [1.+0.j, 2.+0.j, 3.+0.j]

11. Which of the following statement is false?
A. ndarray is also known as the axis array.
B. ndarray.dataitemSize is the buffer containing the actual elements of the array.
C. NumPy main object is the homogeneous multidimensional array
D. In Numpy, dimensions are called axes

12. If a dimension is given as ____ in a reshaping operation, the other dimensions are automatically calculated.
A. Zero
B. One
C. Negative one
D. Infinite

13. What will be output for the following code?

```
import numpy as np
dt = dt = np.dtype('i4')
print dt
```

A. int32
B. int64
C. int128
D. int16

14. Which of the following statement is TRUE?
A. Outliers should be identified and removed always from a dataset.

B. Outliers can never be present in the testing dataset.
C. Outliers is a data point that is significantly close to other data points.
D. The nature of our business problem determines how outliers are used.

15. NumPy is often used along with packages like?
A. Node.js
B. Matplotlib
C. SciPy
D. Both B and C

16. The most important object defined in NumPy is an N-dimensional array type called?
A. ndarray
B. narray
C. nd_array
D. darray

17. Numpy developed by?
A. Guido van Rosum
B. Travis Oliphant
C. Wes McKinney
D. Jim Hugunin

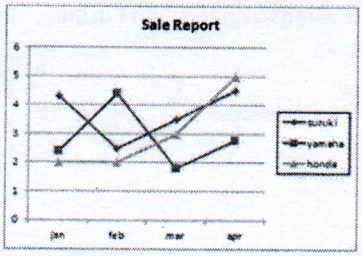
18. Which of the following Numpy operation are correct?
A. Mathematical and logical operations on arrays.
B. Fourier transforms and routines for shape manipulation.
C. Operations related to linear algebra.
D. All of the above

19. The basic ndarray is created using?
A. numpy.array(object, dtype = None, copy = True, subok = False, ndmin = 0)

B. numpy.array(object, dtype = None, copy = True, order = None, subok = False, ndmin = 0)
C. numpy_array(object, dtype = None, copy = True, order = None, subok = False, ndmin = 0)
D. numpy.array(object, dtype = None, copy = True, order = None, ndmin = 0)

20. Which of the following does not visualize data?
A. Charts
B. Maps
C. Shapes
D. Graphs

21. In the given chart, box surrounded with red border is called



A. Data series
B. Chart Title
C. Markers
D. Legend

22. To display histogram with well-defined edge we can write.
A. df.plot(type = 'hist', edge = 'red')
B. df.plot(type = 'hist', edgecolor = 'red')
C. df.plot(type = 'hist', line = 'red')
D. df.plot(type = 'hist', linecolor = 'red')

23. Plot which is used to given statistical summary is
A. Bar
B. Line

C. Histogram
D. Box plot

24. To compare data we can use ____ chart.
A. Line
B. Bar
C. Pie
D. Scatter

25. Which of the following is incorrect regarding Data Visualization?

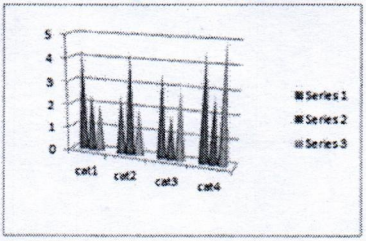
A. Data visualization can be done using Matplotlib library in python.
B. Visualizing large and complex data does not produce effective results.
C. Data visualization is immensely useful in data analysis.
D. Decision makers use data visualization to understand business problems easily and build strategies.

26. Matplotlib is _____ plotting library
A. 1D
B. 2D
C. 3D
D. All of above

27. Data _____ refers to graphical representation of data.
A. Visualization
B. Analysis
C. Plotting
D. Handling

28. consider the code given below:
plt.bar(cities, population, color = ['r','g','b','m'])
what will be color of last bar?
A. Magenta
B. Green
C. Blue
D. Black

29. Following chart is type of



A. Line Chart
B. Bar chart
C. Histogram
D. Scatter chart

30. The interface of Matplotlib used for data visualization is:
A. Seaborn
B. Anaconda
C. Matlab
D. Pyplot

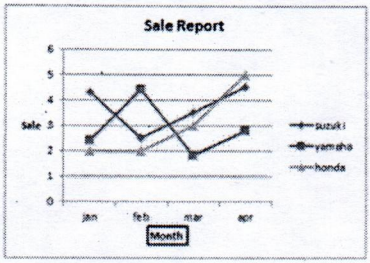
Section B

Note: Attempt all questions. **40 x 0.5 = 20 Marks**

31. The best suitable real life example of data visualization is:

A. Percent of population of by age group in India
B. Google Analytics
C. Solar system drawing
D. Both a and b

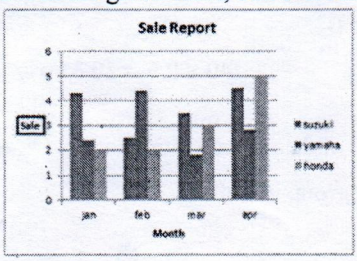
32. in the given chart, box surrounded with red border is called



A. Label
B. X label
C. Y label
D. Legend

33. The command to install Matplotlib library in python is:
A. Install pip matplotlib
B. Install matplotlib
C. Pip matplotlib
D. Pip install matplotlib

34. In the given chart, box surrounded with red border is called:



A. Label
B. X label
C. Y label
D. Legend

35. Select the correct statement to display horizontal box plot
A. Plt.box(data, vert=False)
B. Plt.box(data, horiz=True)

C. Plt.boxplot(data, vert = False)
D. Plt.boxplot(data, horiz = True)

36. Data selection is _____

A. The actual discovery phase of a knowledge discovery process

B. The stage of selecting the right data for a KDD process

C. A subject-oriented integrated time-variant non-volatile collection of data in support of management

D. Record oriented classes finding

37. To remove noise and inconsistent data ____ is needed.

A. Data Cleaning
B. Data Transformation
C. Data Reduction
D. Data Integration

38. Multiple data sources may be combined is called as _____
A. Data Reduction
B. Data Cleaning
C. Data Integration
D. Data Transformation

39. _____ studies the collection, analysis, interpretation or explanation, and presentation of data.
A. Statistics
B. Visualization
C. Data Mining
D. Clustering

40. In real world multidimensional view of data mining, the major dimensions are data, knowledge, technologies, and _____
A. Methods

B. Application
C. Tools
D. Files

41. An _____ is a data field, representing a characteristic or feature of a data object.
A. Method
B. Variable
C. Task
D. Attribute

42. The values of a _____ attribute are symbols or names of things.
A. Ordinal
B. Nominal
C. Ratio
D. Interval

43. "Data about data" is referred to as _____.
A. Information
B. Database
C. Metadata
D. File

44. What are the different types of attributes?
A. Nominal
B. Ordinal
C. Special
D. All of the Above

45. Examples of Nominal can be:
A. ID Numbers, eye colour, zip codes
B. Rankings, taste of potato chips, grades, height
C. Calendar dates, temperatures in Celsius or Fahrenheit, phone numbers
D. The temperature in Kelvin, length, time, counts

46. The type of a Nominal attribute depends on which of the following properties:
A. Distinctness & order
B. Distinctness, order & addition
C. Distinctness
D. All 4 properties

47. Important Characteristics of Structured Data are:
A. Generality
B. Dimensionality
C. Resolution
D. All of the Above

48. What are some examples of data quality problems:
A. Noise and outliers
B. Duplicate data
C. Missing values
D. All of the Above

49. Which Method is used for encoding the categorical variables?
A. LabelEncoder
B. OneHotEncoder
C. CategoryEncoder
D. All of the Above

50. Over fitting happens due to -
A. Imbalance in data
B. Noise in data
C. Data has a high variance
D. All of the Above

51. Which of the following is true about outliers -
A. Data points that deviate a lot from normal observations
B. Can reduce the accuracy of the model
C. Both A and B

D. None

52. Some of the Imputation methods are -
A. Imputation with mean/median
B. Imputing with random numbers
C. Imputing with one
D. All of the above

53. The purpose of feature scaling is to -
A. Accelerating the training time
B. Getting better accuracy
C. Both A and B
D. None

54. In standardization, the features will be rescaled with -
A. Mean 0 and Variance 0
B. Mean 0 and Variance 1
C. Mean 1 and Variance 0
D. Mean 1 and Variance 1

55. Which of the following(s) is/are features scaling techniques?
A. Standardization
B. Normalization
C. Min-Max Scaling
D. All of the Above

56. How to handle the missing values in the dataset?
A. Dropping the missing rows or columns
B. Imputation with mean/median/mode value
C. Taking missing values into a new row or column
D. All of the above

57. The correct way of pre-processing the data should be-
A. Imputation ->feature scaling-> training
B. Feature scaling->imputation->training
C. Feature scaling->label encoding->training
D. None

58. Which one is a feature extraction example?
A. Constructing a bag of words model
B. Imputation of missing values
C. Principal component analysis
D. All of the Above

59. Which of these techniques is used for normalization in text mining?
A. Stemming
B. Stop words removal
C. Lemmatization
D. All of the above

60. When you print/display any series then the left most column is showing value.
A. Index
B. Data
C. Value
D. None of the above

61. _____ is used when data is in Tabular Format.
A. NumPy
B. Pandas
C. Matplotlib
D. All of the above

62. Which of the following command is used to install pandas?
A. pip install pandas
B. install pandas
C. pip pandas
D. None of the above

63. A _____ is a collection of data values and operations that can be applied to that data.
A. Data Structure
B. Data Frame

C. Table
D. None of the above

64. A _____ is a one-dimensional array.
A. Data Frame
B. Series
C. Both of the above
D. None of the above

65. Which of the following function/method help to create Series?
A. series( )
B. Series( )
C. createSeries( )
D. None of the above

66. How many values will be there in array1, if given code is not returning any error?

>>> series4 = pd.Series(array1, index = ["Jan", "Feb", "Mar", "Apr"])

A. 1
B. 2
C. 3
D. 4

67. What type of error is returned by following code?
import pandas as pd
S1 = pd.Series(data = (31, 2, -6), index = [7, 9, 3, 2])
print(S1)
A. SyntaxError
B. IndexError
C. ValueError
D. None of the above

68. We can imagine a Pandas Series as a _____ in a spreadsheet.

A. Column
B. Cell
C. Table
D. None of the above

69. What type of error is returned by following statement?

```
import pandas as pnd
pnd.Series([1,2,3,4], index = ['a','b','c'])
```

A. SyntaxError
B. IndexError
C. ValueError
D. None of the above

70. Which attribute is used to give user defined labels in Series?
A. index
B. data
C. values
D. None of the above

Printed Pages: 04 **University Roll No: .....**

End Term Examination, Even Semester 2021-22
B. Tech. (CS/ME/Auto/MTE/SM), IV & VI Semester
MBAM 0002: Leadership & Organization Behaviour

Time: 1 Hour **Maximum Marks: 100**

Note: Attempt All Questions 50 x 2 = 100 marks

1. Organizational Structure involves:
(a) Policy Statements developed by the firm (b) Resource Allocation
(c) Department Location (d) How activities are coordinated and controlled

2. The _____ leadership style is an expression of the leader's trust in the abilities of his subordinates.
(a) Participative (b) Delegative (c) Authoritarian (d) All of the above

3. Herzberg propounded:
(a) Need Hierarchy (b) Two Factor Theory (c) ERG Theory (d) None of the above

4. Insurance companies usually targets _____ need level of individuals.
(a) Growth (b) Basic (c) Self Actualization (d) Security

5. Process of perception is_____
(a) Receiving-Selecting-Interpreting-Organizing
(b) Receiving-Interpreting-Selecting-Organizing
(c) Receiving-Selecting-Organizing-Interpreting
(d) Organizing-Selecting-Interpreting-Receiving

6. Power gained because of the experience and expertise is _____
(a) Legitimate power (b) Referent power (c) Reward power (d) Expert power

7. Components of Attitude are_____
(a) Affective-Behavioural-Response (b) Concentration-Action-Behavioural
(c) Affective-Behavioural-Cognitive (d) All of the above

8. Mr. Z is being threatened to thrown out from the job if the targets are not being achieved. Which type of power is being exercised here?
(a) Coercive Power (b) Referent Power (c) Reward Power (d) None of the above

9. Power which comes from being trusted and respected is known as
(a) Expert power (b) Legitimate power (c) Referent power (d) Reward power

10. Groups which are formed as the consequence of organisational structure and work division are known as
(a) Informal Groups (b) Formal Groups (c) Target Groups (d) Operational Groups

11. According to Need Hierarchy Theory, in order to motivate an employee there are five basic needs that are meant to be fulfilled step by step. Choose correct order of needs from following
(a) Security, Physiological, Social, Esteem recognition, Self actualization
(b) Esteem recognition, Self actualization, Physiological, Social, Security
(c) Physiological, Security, Social, esteem recognition, Self actualization
(d) Physiological, Social, Security, Self actualization, Esteem recognition

12. A's power over B derives from, and is equals to, B's_____
(a) Dependence on A (b) Power over A (c) Connection with C (d) None of the above

13. Adjourning is the_____ stage in group formation.
(a) First (b) Second (c) Fourth (d) Last
14. Concept of ethical leadership
(a) Exist (b) Doesn't Exist (c) Doesn't Exist in India (d) None of the above
15. In change management, role of education and knowledge
(a) Is important (b) Is very less (c) Can't be checked (d) None of the above
16. Organizational behaviour is
(a) An art (b) Science (c) Social Science (d) All of the above
17. Feature(s) of Maslow's need hierarchy theory is (are):
(A) Theory of human motives (B) Classifies basic human needs in a hierarchy
(C) Theory of human motivation (D) All of the above
18. All the unique traits and patterns of adjustment of the individual is known as
(A) Personality (B) Responsibility (C) Creativity (D) Authority
19. To be efficient every manager must try to
(A) Understand human behaviour (B) Predict human behaviour
(C) Control human behaviour (D) All of the above
20. Organizational objectives can be achieved by
(A) Group work (B) Division of labour (C) Hierarchy of authority (D) All of the above
21. Politics in the organization can be controlled by
(a) Goal Displacement (b) Clarity in job role (c) Saturation in promotion (d) Both (B) and (C)
22. An Ethical Leader
(a) Avoids employee's freedom, growth and development
(b) Supports a nonchalant working environment in the team
(c) is Fair and just in financial matters
(d) Discourages a balanced work life environment
23. Narcissistic Leaders are those who
(a) Worry obsessively about the feeling of others (b) Finds it difficult to confront others
(c) Avoids accepting challenging projects (d) Tries to overcome self doubt with bigger projects
24. In Vroom's Expectancy theory, Expectancy indicates
(a) Probability of achieving that outcome (b) Advancement (c) Award (d) Accounting
25. Which statement is true?
(a) Reward power is based on the leader's ability to provide rewards to those he governs
(b) Traditional authority evolves into rational authority through formal legitimization
(c) Charismatic leaders usually possess magical qualities
(d) All of the above
26. Psychoanalytic theory was propounded by
(a) Allport (b) Henry Murray (c) Sigmund Freud (d) Eysenck
27. Concept of Refreezing states
(a) Avoid old or irrelevant working process of organization
(b) The actual changes related with business practices, staffing or other areas are determined.
(c) The changes are normalized in the organization's day-to-day activities.
(d) Both (b) and (c)

28. Perception is

(a) A subjective process, different people may perceive the same environment differently based on what particular aspects of the situation they choose to selectively absorb, how they organise this information and the manner in which they interpret it to obtain a grasp of the situation

(b) The process through which people select, organise and interpret or attach meaning to events happening in the environment

(c) A much more complex process by which an individual selectively absorbs or assimilates that stimuli in the environment, cognitively organises the perceived information in a specific fashion and then interprets the information to make an assessment about what is going on in one's environment

(d) All of the above

29. In expectancy theory_____ is one of the relationships proposed

(a) Reward-satisfaction relationship (b) Satisfaction-performance relationship

(c) Rewards-personal goal relationship (d) Effort-satisfaction relationship

30. Identify the base of power when a bank robber points a gun at a bank employee.

(a) Referent power (b) Legitimate power (c) Informative power (d) Coercive power

31. Which of the following illustrates that a relatively permanent change in behaviour occurs as a result of experience

(a) Behaviour Modification (b) Learning (c) Motivation (d) Skills

32. The environmental force that shapes personality is known as_____

(a) Gender (b) Height (c) Brain size (d) Experience

33. Which of the following is largely childish, irrational, never satisfied and demanding to others

(a) Super ego (b) Id (c) Ego (d) High ego

34. Which of the following factors must be included in for the success of any effort at improving quality and productivity

(a) Quality management programs (b) Employees participation

(c) Manufacturing simplification (d) Manager's reaction

35. Which stage comes after Norming stage in group formation

(a) Adjourning (b) Storming (c) Performing (d) Forming

36. First impression error can be counted into perceptual errors.

(a) True (b) False (c) Statement is partially correct (d) Can't say

37. Judging an individual on the basis of group characteristics is known as:

(a) Stereotyping (b) Halo Effect (c) Horn Effect (d) None of the above

38. Judging an individual on the basis of only single positive trait is known as

(a) Stereotyping (b) Horn Effect (c) Halo Effect (d) None of the above

39. "How we take the things around us". It is connected with:

(a) Leadership (b) Motivation (c) Perception (d) All of the above

40. "Where we want to go?" In an organizational context primarily it represents:

(a) Mission (b) Values (c) Vision (d) None of the above

41. In Vroom's Expectancy Theory, Valence means

(a) Strength of an individual's preference for a particular outcome (b) Salary

(c) Supervision (d) Knowledge of planning

42. Attitude is
(a) Tendency to react positively (b) Tendency to react negatively
(c) Tendency to react in a certain way (d) All of the above
43. Who has developed expectancy theory
(a) Abraham Maslow (b) Clayton P. Alderfer (c) David McClelland (d) Victor Vroom
44. Your boss never gives you the benefit of the doubt. When you were late back from lunch, he assumed that you had simply taken too much time. He never considered that the elevators were out and you had to walk up 10 flights of stairs. Your boss is guilty of _____.
(a) Self-serving bias (b) Selective perception
(c) Fundamental attribution error (d) Inconsistency
45. In ----------leadership, there is a complete centralization of authority in the leader
(a) Democratic (b) Autocratic (c) Free rein (d) Bureaucratic
46. In---------- in fact "No leadership at all"
(a) Democratic (b) Autocratic (c) Free rein (d) Bureaucratic
47. Free rein leadership is also known as
(a) Democratic (b) Autocratic (c) Laissez-faire (d) Bureaucratic
48. ----------leadership emphasize on rules and regulation in an organization
(a) Democratic (b) Autocratic (c) Laissez-faire (d) Bureaucratic
49. Which of the following fields has most helped us understand differences in fundamental values, attitudes, and behaviour among people in different countries?
(a) Anthropology (b) Psychology (c) Political science (d) Operations research
50. Mr. Sajeev rated Mr. Rajiv high in his job evaluation because both belong to same area and graduated from the same University. It is an example of:
(a) Central Tendency (b) Halo effect (c) Similar-to-me effect (d) Misperception

Printed Pages: 04 University Roll No. .....

End Term Examination, Even Semester 2021-22
B. Tech. CS/CE/ME – IV, VI & VIII Semester
BCSM 0001 : Introduction to Cyber Security

Time: 1 Hour Maximum Marks: 100

Attempt All Questions **2 x 50 = 100 Marks**

1. A hash function guarantees integrity of a message. It guarantees that message has not been_____
A. Replaced B. Over view C. Changed D. Violated

2. In computer security, ...means that the information in a computer system only be accessible for reading by authorized parities.
A. Confidentiality B. Integrity C. Availability D. Accountability

3. The type of threats on the security of a computer system or network are.....
i) Interruption ii) Interception iii) Modification iv) Creation v) Fabrication
A. i, ii, iii and iv only B. ii, iii, iv and v only
C. i, ii, iii and v only D. All i, ii, iii, iv and v

4. Which of the following is independent malicious program that need not any host program?
A. Trap doors B. Trojan horse C. Virus D. Worm

5. The ..... is code that recognizes some special sequence of input or is triggered by being run from a certain user ID of by unlikely sequence of events.
A. Trap doors B. Trojan horse C. Logic Bomb D. Virus

6. The ..... is code embedded in some legitimate program that is set to "explode" when certain conditions are met.
A. Trap doors B. Trojan horse C. Logic Bomb D. Virus

7. The section deals with legal recognition of digital signature
A. Section 3 B. Section 5 C. Section 6 D. Section 4

8. ..... programs can be used to accomplish functions indirectly that an unauthorized user could not accomplish directly.
A. Zombie B. Worm C. Trojan Horses D. Logic Bomb

9. State whether true of false.
i) A worm mails a copy of itself to other systems.
ii) A worm executes a copy of itself on another system.
A. True, False B. False, True C. True, True D. False, False

10. A ..... is a program that can infect other programs by modifying them, the modification includes a copy of the virus program, which can go on to infect other programs.
A. Worm B. Virus C. Zombie D. Trap doors

11. All of the following are examples of real integrity and privacy risks EXCEPT
A. Viruses B. Identity theft C. Hackers D. Spam

12. Which of the following are true about firewalls?
A. Follows a set of rules B. Can be either a hardware or software device
C. Filters network traffic D. All the above

13. Verification of a login name and password is known as
A. configuration B. accessibility C. authentication D. logging in

14. Why is it unethical to share copyrighted files with your friends?
A. It is not unethical, because it is legal
B. It is unethical because the files are being given for free
C. Sharing copyrighted files without permission breaks copyright laws
D. It is not unethical because the files are being given for free

15. The typical computer criminal is a(n):
A. Young hacker
B. Trusted employee with no criminal record
C. Trusted employee with a long, but unknown criminal record
D. Overseas young cracker

16. The common name for the crime of stealing passwords is
A. Jacking B. Identify Theft C. Spoofing D. Hacking

17. Malicious software is known as
A. Badware B. Malware C. Maliciousware D. Illegalware

18. What type of virus uses computer hosts to reproduce itself?
A. Time Bomb B. Worm C. Melissa virus D. Macro Virus

19. The thing that eventually terminates a worm virus is a lack of
A. Memory or disk space B. Time C. CD drive's space D. CD-RW

20. When a logic bomb is activated by a time-related event, it is known as a
A. Time-related bomb sequence B. Virus C. Time Bomb D. Trojan Horse

21. How often will your bank request a confirmation of your password or account number via email?
A. Once a month B. Never C. Once a year D. Once every six months

22. You just received an email from a very wealthy person that needs your help moving money across an international border. For just a few dollars you can help him and he is offering to pay you many times more than your out of pocket expenses. What do you do?
A. Take him up on his offer. Easy money!!!
B. Delete the email. If this happens at work, report the attempt to the IT security office.
C. Forward the message to your friends so they can have a chance to make money as well.
D. Reply with a note that you know it's a scam and are smart enough to recognize the attempt.

23. When browsing online, a new window pops up stating that a virus has been found on your computer. The window provides a button to click offering to resolve the issue. Your best course of action is to:
A. Click on the button to remove the virus.
B. Place your cursor over the button and check the link's website address (URL). If the address looks legitimate, click on it. If it looks like a scam link, close the window.
C. Close both the original browser window and the new "pop-up" window. Do not return to that site.
D. Hit the back

24. What is SSL also referred to as in web browsers?
A. SecureNow B. SMTP C. FTP D. HTTPS

25. Why would a hacker use a proxy server?
A. To create a stronger connection with the target.
B. To create a ghost server on the network.
C. To obtain a remote access connection.
D. To hide malicious activity on the network.

26. Which of the following is not a factor in securing the environment against an attack on security?
A. The education of the attacker
B. The system configuration
C. The network architecture
D. The business strategy of the company

27. To hide information inside a picture, what technology is used?
A. Rootkits B. Bitmapping C. Steganography D. Image Rendering

28. Which of the following is not a typical characteristic of an ethical hacker?
A. Excellent knowledge of Windows.
B. Understands the process of exploiting network vulnerabilities.
C. Patience, persistence and perseverance.
D. Has the highest level of security for the organization.

29. What is short for malicious software (is software designed to disrupt computer operation, gather sensitive information, or gain unauthorized access to computer systems)?
A. Malware B. Moleculewar C. Malisoft D. Malairasoft

30. What is the purpose of a Denial of Service attack?
A. Exploit a weakness in the TCP/IP stack
B. To execute a Trojan on a system
C. To overload a system so it is no longer operational
D. To shutdown services by turning them off

31. What are some of the most common vulnerabilities that exist in a network or system?
A. Changing manufacturer, or recommended, settings of a newly installed application.
B. Additional unused features on commercial software packages.
C. Utilizing open source application code
D. Balancing security concerns with functionality and ease of use of a system.

32. Performing hacking activities with the intent on gaining visibility for an unfair situation is called _____.
A. Cracking B. Analysis C. Hacktivism D. Exploitation
33. Sniffing is used to perform _____ fingerprinting.
A. Passive stack B. Active stack C. Passive banner grabbing D. Scanned
34. Phishing is a form of _____.
A. Spamming B. DOS C. Impersonation D. Scanning
35. Keyloggers are a form of _____.
A. Spyware B. Shoulder surfing C. Trojan D. Social engineering
36. What is the best statement for taking advantage of a weakness in the security of an IT system?
A. Threat B. Risk C. Exploit D. Vulnerability
37. The first computer virus is --------
A. I Love You B. Blaster C. Sasser D. Creeper and Brain
38. MCAfee is an example of
A. Photo Editing Software B. Quick Heal C. Virus D. Antivirus
39. Which of the following is known as Malicious software?
A. illegalware B. badware C. malware D. maliciousware
40. Which of the following is/are threats for eiectronic payment systems?
A. Computer worms B. Computer virus C. Trojan horse D. All of the above
41. Firewalls are used to protect against --------
A. data driven attacks B. fire attacks C. virus attacks D. unauthorized access
42. Which of the following describes programs that can run independently travel from system to system and disrupt computer communication?
A. Viruses B. Trojans C. Droppers D. Worm
43. Authentication is
A. modification B. insertion
C. hard to assure identity of user on a remote system D. none of the above
44. -------- are attempts by individuals to obtain confidential information from you to falsifying their identity.
A. Computer viruses B. Phishing scams C. Phishing trips D. Spyware scams
45. What refers to stealing money or resources a bit at a time
A. Salami Attack B. Identity theft C. Pigeon drop D. Pharming
46. Which of the following attack involves altering data before it is processed by the computer so that incorrect results are obtained.
A. The Salami Attack B. Data Diddling C. DoS D. Web Jacking
47. Which section of IT Act deals with Hacking of computer systems and its penalties?
A. Section 64 B. Section 65 C. Section 66 D. Section 67
48. Which section of IT Act deals with Cyber terrorism?
A. Section 66C B. Section 66B C. Section 66F D. Section 66A
49. Which are the sections of IT Act applicable for Cyber pornography?
A. 66, 66A, 66B B. 67, 67A, 67B C. 68, 68C, 67D D. None of the above
50. Which section of IT Act 2000 propose a punishment of life imprisonment?
A. Section 66F B. Section 66C C. Section 66B D. Section 66A

COMMON Paper

Printed Pages: 04 **University Roll No: .....**

End Term Examination, Even Semester 2021-22
B. Tech, CE, EC, EE, EN, CSE 2nd , 3rd & 4th Year
BCHM 0101: Disaster Management

Time: 1 Hour **Maximum Marks: 100**

Note: Attempt All Questions 50 x 2 = 100 marks

1. The destructive powers of tsunami result mainly from their _____.
   a) incredible height b) unpredictability
   c) cold water d) momentum and long wavelength
2. One of the main reasons for farmers commit suicide in India is
   a) Crop failure b) Earthquake
   c) Tsunami d) All of the above
3. A tiltmeter is a tool used to study which of these natural disaster events?
   a) Tsunami b) Tornado
   c) Hurricane d) Volcano
4. A hypocenter is associated with potentially deadly natural phenomenon?
   a) Earthquake b) Thunderstorm
   c) Hurricane d) Cyclone
5. What does "tsunami" mean in Japanese?
   a) Severe earthquake b) Huge wind
   c) Harbour wave d) Terrible storm
6. Which of the following is most likely to be in the 'eye' of a cyclone?
   a) It is an area of high pressure b) It is an area of low pressure
   c) It has high speed winds d) It has lots of clouds
7. A cyclone is called a_____ in Philippines and Japan.
   a) Hurricane b) Typhoon
   c) Tornado d) Thunderstorm
8. _____is ash from a volcanic mixer with water to form a thick river of mud.
   a) Lahars b) Soilfication
   c) Debris flow d) Creep
9. What is it called when a large number of people in a community get a disease at the same time?
   a) Influx b) Black Death
   c) Epidemic d) Pandemic
10. The Richter scale expresses an earthquakes
    a) Magnitude b) Location
    c) Duration d) Depth
11. Earthquakes occur when there is a sudden release of stored up energy in Earth's
    a) Inner core b) Outer core
    c) Upper mantle d) Lower crust
12. Tsunamis are most commonly found in which ocean?
    a) Pacific Ocean b) Atlantic Ocean
    c) Artic Ocean d) Indian Ocean

13. During a tsunami a series of waves radiating out ward from a central disturbance of the sea floor is called
    a) Ocean storm b) Wind-generated disturbance
    c) Wave train d) Wavelets
14. What does "tsunami" mean in Japanese?
    a) Severe earthquake b) Huge wind
    c) Harbour wave d) Terrible storm
15. Volcanic erupted material when inside the hill/earth/mountain it is called
    a) Lava b) Magma
    c) Lahars d) None of these
16. The National Disaster Management Authority (NDMA) is headed by
    a) Prime Minister of India b) President of India
    c) Governor of States d) Chief Minister of States
17. What is Lahars?
    a) Magma b) Lava
    c) Volcanic mud flow d) None of the above
18. About 2/3rd of the cyclones that occur in the Indian coastline occur in the
    a) Bay of Bengal b) Coastal area of south India
    c) Coastal area of west India d) None of the above
19. The point of the earth's surface directly above the point where an earthquake occurs is called
    a) Focus b) Epicenter
    c) Fracture d) Fault
20. In India National Institute of Disaster Management is located at
    a) Manipur b) Punjab
    c) Hyderabad d) New Delhi
21. Generally the number on Richter Scale ranges between –
    a) 0 and 6 b) 0 and 9
    c) 1 to 5 d) 1 to 12
22. Which of the following is not a man-made hazard?
    a) Leakage of Toxic waste b) Wars and Civil Strife
    c) Drought d) Environmental pollution
23. From where earth quake waves are generated?
    a) Focus b) Epicenter
    c) Solid inner Core d) None of these
24. Richter scale is a
    a) Logarithmic scale b) Calculus scale
    c) Volumetric scale d) Area to vibration ratio scale
25. What are the three phases of disaster management planning?
    a) Preparation, Response and Recovery b) Preparation, Planning and Perception
    c) Evacuating, Rebuilding and Re-branding d) Planning, Evacuating and Recovery
26. A disease that becomes unusually widespread and even global in its reach is referred to as
    a) Epidemic b) Pandemic
    c) Spanish flu d) Hyper endemic

27. The point of the earth's surface directly above the point where an earthquake occurs is called
    a) Focus
    b) Epicenter
    c) Fracture
    d) Fault
28. The center of a cyclone is a calm area and is called the storm.
    a) Eye
    b) Focus
    c) Centre
    d) Radius
29. Ozone layer formation occur in
    a) Troposphere
    b) Stratosphere
    c) Mesosphere
    d) Ionosphere
30. Atmospheric pressure is measured by
    a) Anemometer
    b) Barometer
    c) Colorimeter
    d) Seismometer
31. Which city burned in the "Great Fire" of 1666?
    a) London
    b) Rome
    c) Paris
    d) Berlin
32. The Plan prepared for facilitating the prompt return to the normalcy and continuity of operation after a disaster is called:
    a) Rescue Plan
    b) Relief Plan
    c) Recovery Plan
    d) Response Plan
33. Pareto Analysis' is a:
    a) Risk quantification technique
    b) Risk analysis technique
    c) Risk mitigation technique
    d) Risk prioritization technique
34. Monitoring the levels of exposure to toxic substances is medically done through analysis of body fluids e.g. blood, urine, expired air etc. The process is known as
    a) Condition monitoring
    b) Biological conditioning
    c) Exposure monitoring
    d) Biological monitoring
35. In which Phase of Disaster damage Assessment is done?
    a) Warning phase
    b) Impact Phase
    c) Rescue phase
    d) Re-Habilitation Phase
36. Covid19 is due to a virus named
    a) MERS-CoV
    b) SARS-CoV
    c) MERS-CoV2
    d) SARS-CoV2
37. Why does site history have to be considered for hazard identification?
    a) To estimate the risk
    b) To calculate carcinogenic exposure
    c) To know the probable source and causes of contamination on site
    d) For determination of remedial actions
38. Which of the following data is not required for hazard identification?
    a) Land use
    b) Contaminant levels
    c) Affected population
    d) Estimation of risk
39. Responsibility for securing the scene, preserving life and treating the wounded is the responsibility of
    a) First responders
    b) Mental health professionals
    c) Social workers who specialize in crisis management
    d) All citizens

40. What cloud type can form a tornado?
    a) Cirrus
    b) Cumulus
    c) Cumulonimbus
    d) Stratus
41. Where did the strongest earthquake ever recorded occur?
    a) Japan
    b) USA
    c) Chile
    d) India
42. Which event produces the biggest tsunami?
    a) Earthquake
    b) Underwater landslides
    c) Hurricanes
    d) Impacts of asteroids and comets
43. In 1952, what disaster in London lasted for five days and killed roughly 12,000 people?
    a) Acid rain storm
    b) Polluted fog
    c) River Thames over flowing
    d) Blizzard
44. Vulnerability analysis comes in which part of the Disaster Management Cycle
    a) Mitigation
    b) Preparedness
    c) Response
    d) Recovery
45. In northern hemisphere, the wind of the tropical cyclone blows in
    a) Anticlockwise direction
    b) Clockwise direction
    c) Straight
    d) None of the above
46. About 2/3 rd of the cyclones that occur in the Indian coastline occur in the
    a) Bay of Bengal
    b) Coastal area of south India
    c) Coastal area of west India
    d) None of the above
47. A flash flood is a flood that
    a) Is caused by heavy rain rather than from the flooding of a river
    b) Occurs in urban areas
    c) Occurs suddenly and unexpectedly and for a short duration
    d) Is caused by the blocking of drains.
48. Below average rainfall for a prolonged period of time is called
    a) Famine
    b) Drought
    c) Flood
    d) None of these
49. In which Phase of Disaster damage Assessment is done?
    a) Warning phase
    b) Impact Phase
    c) Rescue phase
    d) Re-Habilitation Phase
50. Which one of the following does not cause Earthquake?
    a) Colliding plates
    b) Coral reefs
    c) Sliding plates
    d) Dividing plates

Printed Pages: 04 **University Roll No. .....**

END Term Examination, Even Semester 2021-22
B.Tech. CSE (CCV, DA, AIML, CSF, IIOT), II Year, IV Semester
BMEM 0002 : Waste To Energy

Time: 1 Hours **Maximum Marks: 100**

Attempt All Questions 2 X 50 = 100 Marks

1. The term ISWM refers to:
a) International Solid Waste Management b) Integrated Solid Waste Management
c) Integrated Solid Waste Machine d) International Solid Waste Mechanism

2. Bio-medical waste can be effectively managed by the thermal process.
a) True b) False

3. Which gas produced in open dumps from the decomposition of biodegradable waste?
a) Ethane b) Methane c) Propene d) Ethene

4. Which of the following cities was the first one to establish a system of waste removal?
(a) Athens (b) Lahore (c) Paris (d) London

5. Which of the following solid wastes describe the term 'Municipal Solid Waste'?
(a) Toxic (b) Hazardous (c) Non-toxic (d) Non-hazardous

6. Why are plastics difficult to recycle?
(a) Because it is a very hard material (b) Because of the different sizes of plastic
(c) Because it is very adhesive in its nature (d) Because of different types of polymer resins

7. Why is recycled paper banned for use in food containers?
(a) Because it creates a lot of spaces
(b) Because it creates contamination
(c) Because paper can be used only one time
(d) Because paper is very thick and can't cover the food containers

8. Which of the following plans are used as a waste management plan?
(a) Use a waste management plan (b) The integrated waste management plan
(c) Recycling of waste management plan (d) Reducing the waste management plan

9. Which of the statements is true for the organic material in the buried solid waste that will decompose?
(a) By the flow of water (b) By the soil particles
(c) By the action of microorganisms (d) By the action of oxidation

10. Which of the following wastes are called the Municipal Solid Waste (MSW)?
(a) Food wastes (b) Wood pieces (c) Plastic cans (d) All of the above

11. The process of burning municipal solid wastes in a properly designed furnace under suitable temperature and these operating conditions is called _____.
(a) Landfill (b) Incineration (c) Recycling (d) Vermicomposting

12. The burning of the wastes is not an acceptable practice of solid waste management because
(a) It is very costly (b) It requires a lot of space
(c) It requires modern technologies (d) It causes several environmental issues

13. Which of the following gas generates when the matter inside the sanitary landfill breaks down?
(a) Methane (b) Nitrogen (c) Hydrogen (d) All of the above

14. Which of the following is the simplest and most common method used in the cities to dump the collected wastes?
(a) River (b) Ocean (c) Landfill (d) None of the above

15. The most disposable wastes are in the form of_____.
(a) Solids (b) Liquids (c) Slurries (d) All of the above

16. Which of the following is true about wastes?
(a) The apparent waste from one process becomes the input to another
(b) All processes of production and consumption produce waste
(c) There is no real waste in nature
(d) All of the above

17. Which of the following methods is a good way of dealing with the solid waste problem?
(a) Recycling (b) Landfilling (c) Both a and b (d) None of the above

18. Which of the following statements is not true for plastic wastes?
(a) Can be used to make compost (b) It lasts for a longer period of time
(c) Toxic fumes are produced when burnt (d) All of the above

19. Which of the following can be recycled many times?
(a) Wood (b) Plastic (c) Aluminium (d) Organic materials

20. Which of the following gas can be produced from landfill wastes?
(a) Biogas (b) Natural gas (c) Liquified petroleum gas (d) All of the above

21. Which of the following statements are the features of zero waste management?
(a) Separate collection of each kind (b) Separation of garbage at the source
(c) Involvement of the community in all activities (d) All of the above

22. How many main components are there in integrated waste management?
(a) Two (b) Three (c) Seven (d) Eleven

23. How many major sources of solid waste are there based on their origin?
a) 10 b) 5 c) 9 d) 6

24. The number of functional components of solid waste management is:
a) 5 b) 3 c) 6 d) 4

25. The Process of Burning of Municipal Solid Waste at High Temperature is called
a) Incineration b) Composting c) Land filing d) none of these

26. Which of the following solid wastes describes the term 'Municipal Solid Waste'?
(a) Toxic (b) Hazardous (c) Non-toxic (d) Non-hazardous

27. Why is it difficult to recycle plastics?
(a) It is very hard (b) It comes in different sizes
(c) It is adhesive (d) It contains different types of polymer resins

28. Which of the following is done on an individual level?
(a) Burning (b) Disposal (c) Recycling (d) Source reduction

29. Why is recycled paper banned for use in food containers?
(a) Because it creates a lot of spaces
(b) Because it creates contamination
(c) Because paper can be used only one time
(d) Because paper is very thick and can't cover the food containers

30. Which of the following plans is used as a waste management plan?
(a) Plan for reuse (b) The integrated plan (c) Plan for recycling (d) Plan for reducing

31. The organic material of the solid waste will decompose
(a) By the flow of water (b) By the soil particles
(c) By the action of microorganisms (d) By oxidation

32. Which of the following wastes is called the Municipal Solid Waste (MSW)?
(a) Food wastes (b) Wood pieces (c) Plastic cans (d) All of the above

33. The process of burning municipal solid wastes under suitable temperature and conditions in a specific furnace is called _____.
(a) Landfill (b) Incineration (c) Recycling (d) Vermicomposting

34. Find the correct statement
(a) The waste from one process becomes the input for another process
(b) All the processes related to consumption and production produce some kind of waste
(c) There is no real waste in nature
(d) All of the above

35. Which of the following statements is incorrect for plastic wastes?
(a) It is used to make compost (b) It lasts for a longer period of time
(c) Toxic fumes are produced when burnt (d) All of the above

36. Identify a Sort of Plan That We Should Make to the Disposal of Solid Waste from the Given Ones?
a) The integrated waste management plan b) Recycling of waste management plan
c) Reducing the waste management plan d) Use of waste management plan

37. Why Do Plastics Fall Under a Difficult Material to Recycle?
a) Because it is a very hard material

b) Because plastic is very adhesive in its nature
c) Because of different types of polymer resins
d) Because of the different sizes of plastic

38. Why is Recycled Paper Banned for Usage in Food Containers?
a) Because paper is used only one time
b) Because it creates contamination
c) Because the paper is very thick and it cannot cover the food containers
d) Because it creates a lot of spaces

39. Identify the Correct One from the Given List About Wastes?
a) There is no real waste in nature
b) The apparent wastes collected from one process becomes the input to another
c) All processes of consumption and production produce waste
d) All of the above

40. Which of the Following Methods is a Good One in Dealing With the Solid Waste Problem?
a) Landfilling b) Recycling c) Both a and b d) None of the above

41.In which method of disposal of MSW, the waste is dumped in soil.
a) Composting b) Landfilling c) both a and b d) none

42. The density of ash produced in MSW is
a) 100 kg/m3 b) 400 kg/m3 c) 700 kg/m3 d) 900 kg/m3

43. The waste produced in cotton mills is
a) MSW b) Hazardous waste c) Non- hazardous waste d) solid waste

44. which of following waste decomposed by bacteria?
a) Ashes b) rubbish c) food waste d) radioactive substance

45. which of the biological method of disposal of MSW?
a) Landfills b) Composting c) shredding d) none of these

46. which of the following is not a result of mishandling of solid waste?
a) Odor b) Hydrolysis of sugar c) fire d) All of the above

47. which of following method cannot be used for waste disposal?
a) Dumping b) burning c) gasification d) open sea disposal

48. Which of the following temperature is optimum for incinerators?
a) 900-1100^{0}C b) 100- 110^{0}C c) 90-110^{0}C d) 100-11000^{0}C

49. why mismanagement of battery waste can be harmful?
a) Heavy metals b) Plastics c) odor d) none of these

50. Which of the following is not a biomedical waste?
a) Animal waste b) Microbiological waste c) Chemical waste d) Domestic waste